

Implementing an Automated Student Records Management System to Enhance Operational Efficiency at Trece Martires City College

A Capstone Project Presented to the Faculty of the Department of Information Science and
Technology School of Sciences and Engineering,
University of Asia and the Pacific
Pearl Drive, Ortigas Center, Pasig City, Metro Manila



In Partial Fulfillment of the Requirements for the Degree of Bachelor
of Science in Information Technology

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ACKNOWLEDGEMENT

Chapter 1

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

Trece Martires City College is a local government-funded higher education institution offering undergraduate programs to approximately 3,000 to 4,000 students. The college employs around 120 to 150 personnel across academic and administrative offices that support student services and institutional operations.

Educational institutions rely heavily on accurate and efficient records management to ensure smooth administrative operations, compliance with legal standards, and quality service delivery to students. For TMCC, maintaining student records is currently carried out through a semi-manual hybrid system that combines physical archiving with basic digital tracking. In this setup, original legal documents—such as enrollment forms, diplomas, and other official papers—are preserved in individual folders and stored alphabetically in filing cabinets. Alongside this physical storage, a simple Excel spreadsheet is utilized to log essential student information including full name, student identification number, academic program, and year level. This spreadsheet serves primarily as a reference index to assist staff in locating the corresponding physical file (Agomor & Test, 2023).

Despite being functional and straightforward to establish, this operational framework involves labor-intensive workflows. Creating a new student record requires staff to manually assemble physical folders and encode data into the spreadsheet. Retrieving information necessitates searching through the Excel file to identify the folder's location before physically retrieving it from storage. Verification processes require manual cross-referencing between the digital index and the hard-copy documents, followed by official signing and sealing to authenticate the records. Furthermore,

whenever updates are needed, they must be applied separately to both the spreadsheet and the physical files, creating redundant work and increasing the likelihood of discrepancies (Quilla et al., 2024).

These inherent limitations present significant operational challenges. The requirement to maintain two separate sources of information heightens the risk of data inconsistency and human error. Manual processes are time-consuming and reduce overall operational efficiency, which can result in delays when responding to inquiries or fulfilling requests. Additionally, as student enrollment increases over time, dependence on physical storage and basic spreadsheet software makes scaling operations difficult and unsustainable (Padilla et al., 2025).

To address these concerns, this study proposes the development and implementation of an Automated Student Records Management System (ASRMS). By shifting toward a centralized digital approach, the proposed system eliminates reliance on spreadsheets and streamlines administrative tasks. Role-based access ensures that authorized Registrar personnel maintain control over student records while administrators are limited to system governance, monitoring, and access control. Rather than replacing physical storage entirely, the system functions as an advanced searchable index that logs the location of physical documents, thereby simplifying retrieval and reducing processing time. Through these features, the ASRMS is designed to reduce errors, improve efficiency, and provide a scalable framework capable of accommodating institutional growth (Furqony et al., 2025).

1.2 Statement of the Problem

The primary problem of this study is the lack of a centralized and real-time automated system at Trece Martires City College for managing student records. The institution currently relies on semi-manual procedures and a hybrid setup that includes physical files and basic digital tools such as spreadsheets and local storage. These processes often lead to operational inefficiencies, data redundancy, and difficulties in maintaining accurate and consistent student records across multiple sources and formats.

This study determines the current status of the existing system by examining how student data is stored, retrieved, maintained, and updated. It also analyzes how records are verified and accessed by administrative personnel. The study covers the manual encoding of data, as well as the handling and organization of both physical and digital files. In addition, it assesses the time and effort required by these manual processes and identifies areas where errors and inconsistencies are likely to occur.

Furthermore, the study identifies key gaps and limitations in the current setup. These include the absence of a centralized database, which hinders efficient management and updating of records. There is also limited accessibility of records for both staff and students. The lack of digital access to student information reduces convenience and transparency, and it increases the workload of registrar office personnel.

To address these issues, this study proposes the design and development of an Automated Student Records Management System. The system will provide a centralized and structured repository for student data. It will implement role-based access control, allowing registrar personnel to perform full Create, Read, Update, and Delete (CRUD) operations securely and efficiently. Students will also be able to access and view their personal and academic information in a convenient and controlled

manner. In addition, they can submit requests for updates, which will be subject to approval by authorized Registrar personnel to ensure data integrity and security. Finally, the study evaluates the level of user acceptance of the developed system among students and registrar personnel. The evaluation is based on ISO 25010 standards, covering functional suitability, usability, reliability, and security. This assessment aims to determine the system's effectiveness and its suitability for adoption in the college.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to implement an Automated Student Records Management System for efficient handling of student records.

1.3.2 Specific Objectives

- a) Assess the existing student records management process at Trece Martires City College in terms of submission procedures, approval workflows, and turnaround time for document requests, identifying limitations in efficiency and accessibility.
- b) Determine the operational challenges of the current system, including manual processing, data redundancy, record inconsistencies, and delays in retrieving student information.
- c) Design and develop an enhanced Automated Student Records Management System (ASRMS) that incorporates role-based access control for students, registrar staff, and administrators, ensuring secure, centralized, and efficient data management.
- d) Implement key system functionalities such as digital submission of personal information update requests within institutional networks, status monitoring, authorized Registrar access for viewing, updating, and managing student records, and controlled student access for viewing approved academic information to streamline transactions and reduce processing time.
- e) Evaluate the developed system's performance and usability based on ISO/IEC 25010 standards, focusing on functional suitability, reliability, usability, and security as experienced by students and registrar personnel.

1.4 Scope and Limitations

The scope of the system includes the automation and centralized management of student records and registrar-related workflows, allowing authorized Registrar personnel to efficiently perform Create, Read, Update, and Delete (CRUD) operations on academic data. Existing student records will be manually encoded, validated, and maintained exclusively by authorized Registrar Office personnel during the initial implementation phase. Administrators are limited only to system administration functions such as user management, monitoring, and security configuration, and are not permitted to perform CRUD operations on student academic records.

The research involves the participation of Registrar Office personnel and a selected group of students during the second semester of Academic Year 2025–2026 to observe active transactions and evaluate system performance. For the initial implementation of the proposed Automated Student Records Management System (ASRMS), only currently active TMCC students for Academic Year 2026–2027 will be included in the system database. Alumni records and previously archived historical records are excluded from the initial implementation scope. However, if the system evaluation after implementation is deemed successful and approved by the institution, future system expansion may include the integration and digitization of older historical and alumni records.

The technical development follows the System Development Life Cycle (SDLC) framework, encompassing requirements analysis, system architecture design using role-based access control (Student, Registrar Staff, and Admin), and the implementation of automated record management

logic. The system is designed to handle academic and personal records, providing centralized and LAN-based access for authorized Registrar personnel to manage records. Students are permitted to view their academic information only after it has been processed and approved by the Registrar. Students are not permitted to directly edit official academic records within the system. They are only authorized to view approved academic information, submit correction or update requests for personal information subject to Registrar approval, request official academic documents, and monitor the status of submitted requests.

The system is primarily designed for deployment within the institution's internal Local Area Network (LAN) and campus-based computer facilities. Authorized Registrar personnel may access the system through institutional computers connected to the internal network, even without public internet access. Students may view their academic records only through designated institutional access points, such as Registrar Office terminals or authorized campus computer stations. The system is not accessible from off-campus locations or through the public internet, and it does not support external cloud deployment or mobile application integration.

Evaluation of the system is delimited to the ISO 25010 standards, focusing on functional suitability, usability, reliability, and security through User Acceptance Testing (UAT).

However, this study has certain limitations. The system is intended solely to assist the administrative operations of the Registrar's Office and does not cover other departments such as Finance, Human Resources, or the Learning Management System (LMS). The system excludes disciplinary records, finance and payment-related records, and LMS-related functions or academic learning activities. The system is designed to work within the current IT infrastructure of the institution and does not utilize mobile application features. It does not serve as the legal repository of

records, as physical documents remain the institution's official files in accordance with existing policies and archival procedures.

The study excludes the digitization of historical or previously archived student records during the initial implementation phase. Only active student records and newly processed academic records within the implementation period will be encoded and managed within the system, while older physical records will remain stored under the institution's existing archival procedures. Initial data entry, record encoding, validation, and maintenance shall be performed manually by authorized Registrar Office personnel. These personnel are responsible for verifying submitted academic documents, ensuring the accuracy of student information, and entering validated records into the centralized system database.

Furthermore, the study is conducted within a specific academic term and institutional environment and does not cover long-term maintenance or integration with external government or national databases.

1.5 Significance of the Study

This study provides a technical and operational framework to address inefficiencies in student records management at Trece Martires City College. By implementing an automated centralized system for managing student records, the research offers the following benefits to its stakeholders:

Students can conveniently view their academic and personal records without repeated physical inquiries, improving accessibility and satisfaction with registrar services. Students may submit

requests to correct or update personal information, which remain subject to Registrar review and approval before any official changes are applied. This ensures accessibility while preserving data integrity and institutional control.

For the Registrar Staff, the research introduces a digitized workflow that eliminates repetitive manual tasks such as encoding data into spreadsheets, maintaining physical folders, and manual verification of student records. Staff can efficiently perform Create, Read, Update, and Delete (CRUD) operations in a centralized database, ensuring data accuracy, reducing redundancy, and streamlining record maintenance. This centralized approach also allows for organized and systematic storage of all student records, improving overall operational efficiency.

For the College Administration, the system supports institutional digital transformation by providing a centralized platform for storing, managing, and maintaining student records securely. It generates accurate logs of database operations and operational summaries, which are essential for data-driven decision-making, quality assurance, and auditing purposes. This shift to automation reduces administrative overhead associated with paper-based tracking and manual data reconciliation.

For Future Researchers, this study serves as a technical reference for developing centralized record management systems in a local government-funded higher education institution. It provides insights into designing systems that prioritize data integrity, secure storage, and role-based access control, offering a foundation for future studies on system scalability, integration with other institutional databases, and advanced data management solutions in higher education.

Chapter 2

REVIEW OF RELATED LITERATURE

Existing research in higher education consistently emphasizes that implementing a centralized Automated Student Records Management System is vital for achieving measurable operational efficiency. Transitioning toward digital frameworks allows institutions to minimize human errors while significantly reducing the turnaround time required for administrative transactions. According to Mugambwa et al. (2025), an automated platform enhances institutional capacity to manage extensive datasets without necessitating an increase in staffing requirements. This efficiency is most evident when manual tasks such as data encoding and verification—common in hybrid operational systems—are transitioned into streamlined record management assistance. Literature indicates that streamlining these processes directly correlates with improved service delivery and more robust institutional governance (Garrison & Vaughan, 2021). By adopting centralized system architectures, academic institutions can ensure a seamless, real-time data flow that eliminates the redundancies associated with manual indexing (Sposato et al., 2023).

Within the Philippine higher education landscape, recent local research underscores the imperative to resolve administrative bottlenecks inherent in conventional manual record-keeping. Duruin (2024) observed that the degradation of physical archives and protracted retrieval intervals serve as primary drivers of dissatisfaction among students in provincial institutions. Local researchers contend that customized automated frameworks demonstrate greater efficacy than generic Excel templates by aligning more closely with institutional policies and specific grading architectures (Reyes et al., 2023). Automating academic trajectories and request tracking ensures that data remains synchronized and accessible for real-time administrative verification, effectively mitigating the

"double-work" challenges prevalent in local colleges. Furthermore, the adoption of centralized digital portals has been demonstrated to significantly reduce the turnaround time for academic record verification (Cabero-Almenara et al., 2024).

Existing research contends that the efficacy of a student records management framework is fundamentally contingent upon robust data security and the implementation of centralized access controls (Valeriano, 2024). Shifting from conventional physical archives toward a digitized, automated operational environment effectively mitigates the vulnerabilities associated with record loss, whether through material degradation, misfiling, or external environmental hazards. The integration of a secure database architecture ensures that sensitive student information remains shielded from unauthorized access through the enforcement of strict, role-based authentication protocols. Furthermore, the utilization of automated audit trails empowers the Registrar's Office to monitor every specific modification within a student's academic profile, establishing a degree of institutional accountability that remains unachievable within fragmented spreadsheet systems. Such transparency is inherently essential for upholding the integrity and credibility of institutional records during rigorous external quality assurance evaluations and administrative audits (World Bank, 2022).

Recent evaluations of digital transitions by Bautista et al. (2025) indicate that the success of automation depends heavily on high user acceptance levels. Their findings suggest that even in resource-limited settings, automated record-keeping can reduce the mental workload of administrative employees by nearly 40%. This reduction in administrative burden allows personnel to focus on high-value tasks such as personalized student support rather than manual filing (Hodges et al., 2021). For the Registrar's staff at Trece Martires City College, shifting away from repetitive manual encoding is key to stabilizing internal workflows and improving productivity. When administrative users feel

empowered by the system's usability, the project's overall effectiveness increases across the institution.

The synthesis of global and local literature reinforces the need for an Automated Student Records Management System (ASRMS) to replace fragmented manual procedures. Current manual workflows involving basic spreadsheets and physical folders are prone to discrepancies and cause significant delays in the release of academic records. This study addresses the gap between existing hybrid capabilities and the operational reality of growing local colleges in the Philippines. By automating CRUD operations and centralizing data, the proposed system strengthens data-driven decision-making for school administrators. The research serves as a catalyst for institutional modernization, ultimately improving the speed and accuracy of academic services to support institutional quality standards (Preprints, 2026).

2.1 Process Automation

The reliance on manual and fragmented record-keeping systems at institutions like Trece Martires City College creates operational inefficiencies that affect the processing and management of student records. Manual methods such as spreadsheets and physical files often lead to data duplication, inconsistent records, and delays in retrieving student information. These challenges increase the workload of Registrar staff, who are required to perform repetitive validation and data entry tasks, reducing overall productivity and efficiency. As a result, processing requests for academic records becomes slower and more prone to human error, particularly in tracking and updating student information.

In addition, the absence of an automated system limits the ability to maintain accurate and real-time records, making it difficult for administrators to monitor system activities and ensure data integrity. Without a centralized system, managing CRUD operations becomes inefficient and less

secure, increasing the risk of inconsistencies in student data. Addressing these issues through an Automated Student Records Management System (ASRMS) helps streamline record handling, reduce administrative workload, minimize errors, and improve the overall efficiency and reliability of student records management.

2.2 Operational Efficiency

The effectiveness of an automated system is defined by its capacity to improve the accuracy, speed, and reliability of institutional transactions. Modern academic platforms are measured by how well they reduce administrative workload while maintaining high data integrity. Studies show that an Automated Student Records Management System (ASRMS) allows for a seamless flow of data by replacing isolated Excel files with a unified database framework. By implementing a centralized architecture, the college can ensure that record retrieval and student certifications are performed with minimal human intervention. Mugambwa et al. (2025) suggest that successful implementation must include a measurable increase in both staff productivity and student satisfaction. Furthermore, the system's effectiveness is supported by secure audit trails that provide transparency in the handling of sensitive student files, replacing the lack of oversight inherent in simple spreadsheets.

The effectiveness of digital records management is also deeply tied to the level of data security and access control implemented. Transitioning to a secure automated environment virtually eliminates the risk of losing vital student transcripts due to physical damage or misfiling. Implementing a secure database ensures that sensitive student information is shielded from unauthorized access through encrypted authentication protocols and role-based permissions. By transitioning from fragmented spreadsheets to a centralized system, the institution can enforce a unified security policy where access is granted only to authorized personnel, such as registrar staff and

administrative heads. These security measures are essential for maintaining the credibility of academic records during external institutional audits or quality assurance evaluations.

Table 1: Synthesis of Literature and Institutional Gaps

CORE THEME	KEY FEATURES	TMCC CURRENT STATUS / GAP
Data Centralization	Transition from isolated digital files to a unified database for a “Single Source of Truth” (Gunawan & Wijaya, 2023).	Student records are fragmented across physical folders and multiple non-synced Excel spreadsheets.
Operational Efficiency	40% reduction in mental workload through intuitive interfaces (Bautista et al., 2025).	High clerical burden and slow manual document retrieval times.
Information Accessibility	Secure viewing of their academic profiles, grades, and personal details through the institutional system dashboard (Cabero-Almenara et al., 2024).	Students must physically visit the Registrar’s Office to verify or view their basic academic information.
Records Management	Centralized databases with encryption and audit trails (Valeriano, 2024).	Fragmented spreadsheets vulnerable to data inconsistency and unauthorized access.

Data Security & Integrity	Centralized databases with role-based access control, encryption, and audit trails (Valeriano, 2024).	Current digital records (Excel) lack audit logs and are vulnerable to accidental deletion or unauthorized access.
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The table above highlights the critical discrepancies between existing administrative standards and the current operational reality at Trece Martires City College. While data centralization and operational efficiency are well-documented as key drivers for institutional success, TMCC remains hindered by a "double-entry" manual workflow and a high clerical workload. The specific gaps in Information Accessibility show that the lack of a secure digital viewing therefore causes students to rely on physical visits for basic record verification, leading to unnecessary foot traffic and administrative delays. Furthermore, the current state of records management relies on fragmented, non-synced spreadsheets which lack the security protocols and auditability recommended in recent academic literature.

These identified gaps necessitate a transition toward a fully Automated Student Records Management System (ASRMS) that streamlines the Registrar's core administrative tasks. By focusing on a centralized database with robust CRUD (Create, Read, Update, Delete) functionalities and a dedicated Student View, the proposed system aims to eliminate the risks associated with manual indexing. This shift ensures that student data is managed through a "Single Source of Truth," providing a scalable roadmap for the college's administrative modernization and data integrity.

2.3 Records Management

Automated records systems centralize student information to offer secure storage, rapid retrieval, and comprehensive audit trails for academic institutions. Leading research suggests that the effectiveness of these platforms is measured by their ability to provide Role-Based Access Control (RBAC) and robust data encryption, ensuring that only authorized personnel can modify sensitive information (Valeriano, 2024). A key feature of modern systems is the transition from static, fragmented files to a dynamic, centralized database where academic data can be managed through CRUD (Create, Read, Update, and Delete) operations in real-time. At Trece Martires City College, academic records are currently fragmented across various physical folders and spreadsheets, a setup that necessitates a time-consuming manual indexing process. This fragmentation often leads to verification errors and data inconsistencies between the physical archives and the digital index. By implementing an automated records management solution, the college can eliminate the need for redundant "double-entry" workflows and establish a "Single Source of Truth." This system not only improves operational efficiency but also provides a dedicated Student Viewing portal, allowing students to access their academic profiles without physical intervention. Ultimately, a modern records system supports institutional governance by providing transparent logs of all data modifications, ensuring the long-term integrity of student records (World Bank, 2022).

2.4 Summary

The synthesis of existing literature confirms that automated systems consistently improve record accuracy and coordination between academic and administrative offices within higher education.

Research indicates that colleges operating on manual or inconsistent platforms—similar to the hybrid setup at TMCC—experience slower transactions, duplicated data entry, and higher error rates (Mugambwa et al., 2025). Studies show that once institutions adopt centralized records management, documentation workloads decrease significantly because data is validated and stored in a single, reliable environment rather than fragmented across multiple files. Findings from recent academic reports suggest that institutional efficiency peaks when students are provided with secure, LAN-based access to view their academic profiles, reducing the administrative burden on registrar personnel (Preprints, 2026).

These research findings directly relate to the current situation at Trece Martires City College, which still relies on fragmented spreadsheets and manual physical validation processes. The literature shows that these challenges are common in growing institutions and that service delays are significantly reduced when the record management lifecycle is digitized through a centralized database (Bautista et al., 2025). Research also notes that automation is most effective when designed specifically for an institution's internal workflows—such as the CRUD (Create, Read, Update, Delete) operations and QR-coded verification methods explored in this study—rather than using generic software templates. Ultimately, the synthesis of global and local literature provides a solid foundation for implementing an Automated Student Records Management System (ASRMS) to achieve total operational modernization and data integrity at the college.

2.5 Definition of Terms

This section provides clear definitions of key terms used in the study to avoid ambiguity and ensure a common understanding among readers. Each term is defined conceptually, based on existing academic literature, and operationally, reflecting how it is specifically applied within the context of this research on the automation and operational efficiency of Trece Martires City College's registrar processes.

ASRMS (Automated Student Records Management System). Conceptually, a chronological record providing evidence of system activities. Operationally, it refers to the system's ability to log every modification (addition, update, or deletion) made to a student's record by registrar personnel.

Centralized Database. Conceptually, a storage mechanism where data is maintained in a single, unified location. Operationally, it refers to the secure SQL-based storage for TMCC student records that replaces non-synced Excel files and physical indexing.

CRUD (Create, Read, Update, Delete). Conceptually, the four basic functions of persistent storage. Operationally, it refers to the primary administrative tools used by the TMCC Registrar to manage student academic history and profiles within the system.

Data Integrity. Conceptually, the accuracy and consistency of information throughout its lifecycle. Operationally, it refers to ensuring that the digital records in the ASRMS remain consistent with verified physical documents and are protected from unauthorized changes.

Operational Efficiency. Conceptually, the degree to which an institution achieves objectives with minimal waste. Operationally, it refers to the measurable reduction in "double-entry" work and the increased speed of record retrieval at the TMCC Registrar.

Process Automation. Conceptually, the use of technology to perform repetitive tasks. In this study, it refers to the digitization of manual registrar workflows, specifically the automated syncing of student data and the generation of QR-coded verification for records.

QR-coded Verification. Conceptually, a machine-readable code used for data authentication. Operationally, it refers to the unique code embedded in student record summaries that allows registrar personnel to instantly verify the authenticity of a digital or printed record.

Role-Based Access Control (RBAC). Conceptually, an approach to restricting system access to authorized users. Operationally, it refers to the security layer where Registrar have full CRUD access while Students are restricted to "View-Only" permissions.

Student Viewing. Conceptually, a user interface for accessing personal information. Operationally, it refers to the secure LAN-based dashboard where TMCC students can log in to view their academic profiles, grades, and personal details in real-time or in an internal system dashboard whenever new documents are available.

User Acceptance Testing (UAT). Conceptually, a phase where end-users verify that a system meets requirements. In this study, it refers to the evaluation where TMCC Registrar personnel and students test the system's usability based on ISO 25010 standards.

Chapter 3

METHODOLOGY

This project addresses operational inefficiencies at Trece Martires City College (TMCC) caused by fragmented data tracking and the “double-entry” burden of a semi-manual records system. The study aims to develop the Automated Student Records Management System (ASRMS) to digitize and consolidate the management of academic data into a single, unified database. By transitioning from isolated spreadsheets to a centralized system, the ASRMS is designed to improve data integrity, eliminate redundant clerical tasks, and provide students with digital access to their academic profiles within institutional networks.

Following the System Development Life Cycle (SDLC), the project ensures that each phase—from requirement analysis and database design to coding, testing, and deployment—aligns with the institutional goal of enhancing operational efficiency. Authorized registrar staff are responsible for manually encoding, validating, updating, and approving student records in the system using authorized registrar staff account interfaces. Students are only permitted to view approved academic records and request updates to personal information subject to registrar approval. The resulting system provides a structured digital solution focused on CRUD (Create, Read, Update, Delete) functionalities and role-based viewing portals, thereby strengthening transparency and administrative coordination at Trece Martires City College

3.1. Conceptual Framework

The conceptual framework of this study follows the Input-Process-Output (IPO) model, integrated with a continuous evaluation phase, to illustrate the systematic modernization of student records management at Trece Martires City College. The framework highlights how the transition from a fragmented semi-manual setup to a centralized automated system addresses operational

bottlenecks—specifically the "double-entry" clerical burden and the lack of digital accessibility for students. By focusing on robust database management and role-based viewing portals, the framework demonstrates a clear pathway toward institutional data integrity and administrative efficiency.

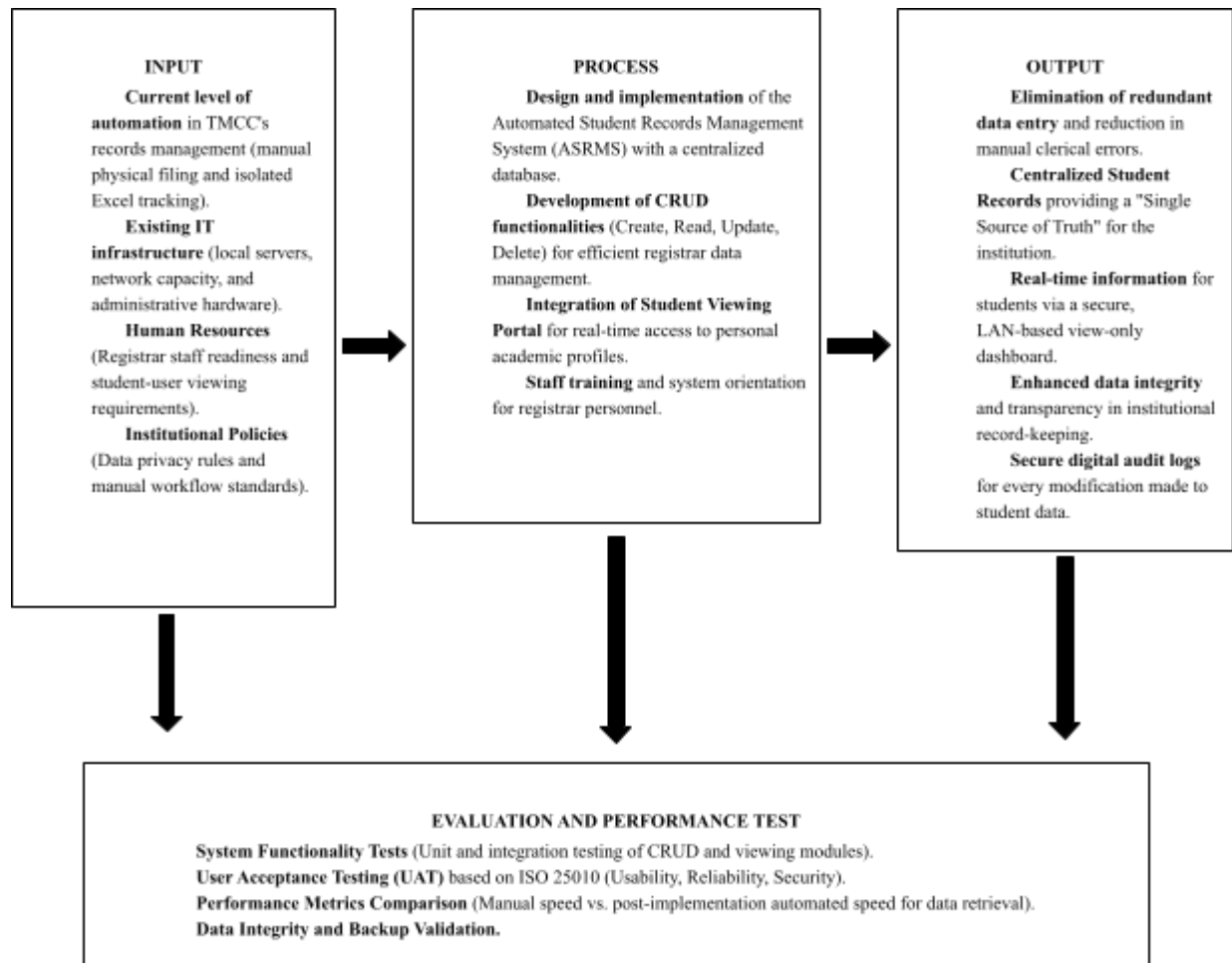


Figure 1: Conceptual Framework of Student Records Management System

The Input phase captures the baseline conditions at Trece Martires City College. This includes the current "semi-manual" state of record-keeping, characterized by fragmented Excel files and physical folder archives. It also considers the existing IT infrastructure, such as local secure servers and network capacity, alongside institutional policies and data privacy rules that define the legal boundaries for handling student academic information.

The Process phase represents the technical development and implementation of the Automated Student Records Management System (ASRMS). Rather than traditional manual indexing, this involves the development of a centralized database with robust CRUD (Create, Read, Update, and Delete) functionalities. This allows the Registrar to manage records in a unified digital environment. Additionally, a dedicated Student Viewing is integrated to provide LAN-based viewing to personal and academic records. Staff training is conducted to ensure registrar personnel can transition from manual filing to digital data management effectively.

The Output phase reflects the immediate results of the transition to automation. Key outcomes include the elimination of redundant "double-entry" workflows and a significant reduction in manual clerical errors. The phase results in a "Single Source of Truth" for all student records, ensuring data consistency across the institution. Furthermore, it provides students with instant, secure access to view their academic profiles without the need for physical office visits. Secure digital audit logs are generated for every modification, enhancing institutional transparency and accountability.

The Evaluation phase determines the system's effectiveness through rigorous functionality tests and User Acceptance Testing (UAT) based on ISO 25010 standards (Usability, Reliability, and Security). The study measures performance metrics, specifically comparing the speed of data retrieval and the accuracy of records against the previous manual baseline. Feedback from this phase provides the necessary data for the continuous refinement of the system's operational efficiency and long-term sustainability.

3.2. Analysis Alternatives

To address the operational inefficiencies at Trece Martires City College (TMCC), several alternatives

for improving the document request and records management processes were considered. Each alternative was evaluated based on feasibility, cost, scalability, and impact on operational efficiency. A decision matrix was used to compare these options, with scores assigned on a scale of 1 to 5 (1 = Very Poor, 5 = Excellent).

Alternative 1: Full Manual Process Optimization

This approach involves improving existing manual procedures through standardization, training, and workflow documentation without implementing new technology.

Pros: Zero software development cost; minimal infrastructure changes.

Cons: Does not solve physical file degradation; maintains high human error rates; fails to provide digital accessibility for students.

Alternative 2: Automated Student Records Management System (ASRMS)

This solution involves a centralized digital system for student records, enabling staff to efficiently perform CRUD (Create, Read, Update, Delete) operations, while students have secure view-only access to their academic profiles.

Pros: Eliminates manual duplication; provides real-time viewing for students within institutional networks; feasible for single-semester implementation.

Cons: Requires staff training; depends on ICT infrastructure.

Alternative 3: Integrated Institutional ERP System

A comprehensive system automating all college departments (Finance, HR, Enrollment) alongside records management.

Pros: Total institutional coverage; eliminates data silos across all departments.

Cons: Prohibitively high initial cost; implementation time exceeds the project’s one-semester timeline.

The system alternatives were evaluated and compared using a decision matrix based on the following criteria: operational efficiency, technical feasibility, cost-effectiveness, implementation speed, and data integrity. Each criterion was rated using a scale of 1 to 5, where 5 represents excellent performance and 1 represents poor performance. The scores were assigned based on system analysis, comparison of features, and how well each alternative meets the needs of Trece Martires City College.

Table 2: Scoring Scale

Description	Scale
Excellent	5
Very Good	4
Good	3
Fair	2
Poor	1

Table 3: Decision Matrix for System Alternatives

Criteria	Alt 1: Manual	Alt 2: ASRMS (Proposed)	Alt 3: Institutional ERP
Operational	2	5	5

Efficiency			
Technical Feasibility	5	5	2
Cost-Effectiveness	4	5	1
Implementation Speed	5	4	1
Data Integrity	1	5	5
TOTAL SCORE	17	24	14

The results of the decision matrix show that the ASRMS (Proposed System) is the most suitable alternative among the three options evaluated. It obtained the highest total score of 24, outperforming both the Manual system with 17 and the Institutional ERP system with 14.

This indicates that ASRMS provides the best overall balance across all criteria. It demonstrates high operational efficiency, strong technical feasibility, excellent cost-effectiveness, and superior data integrity, while still maintaining a relatively fast implementation speed. Compared to the Manual system, ASRMS significantly improves accuracy and efficiency, and reduces human errors. Meanwhile, compared to the Institutional ERP system, it is more practical and affordable to implement within the institution's current resources.

Overall, the results suggest that ASRMS is the most optimal solution because it effectively meets the institution's needs in terms of performance, reliability, and feasibility, without the high cost and

complexity associated with full-scale ERP systems.

3.3. Population Sampling

The population of the study consists of administrators, faculty members, and selected students directly involved in the college's academic record transactions. A purposive sampling technique will be employed, specifically targeting respondents who are primary stakeholders in the institution's information and record-handling processes. This sample includes approximately 10 administrative staff (primarily from the Registrar and Admissions offices), 15 faculty members responsible for grade submission, and 9 student representatives who frequently utilize record services, for a total of 34 respondents. This distribution ensures a balanced perspective across all operational levels of record management.

The insights obtained from this population sample directly informed the ASRMS design specifications, ensuring the system's modules—such as Verification, Request Processing, and Analytics—address actual institutional bottlenecks. By focusing on these specific users, the study ensures that the evaluation of "operational efficiency" is grounded in first-hand experience. This data-driven approach strengthens the system's validity and relevance, providing a solid foundation for the successful development and implementation of TMCC's automated student records management system.

3.4. Research Instruments

The research will utilize several instruments to collect accurate and comprehensive data essential for implementing the Automated Student Record Management System (ASRMS) for Trece Martires City College. These instruments are designed to gather both quantitative and qualitative

data, ensuring a holistic understanding of institutional record-handling challenges and specific automation needs.

The main research instrument is a structured questionnaire, validated by experts in information systems and educational management. Using a five-point Likert scale, the questionnaire evaluates the level of process automation and operational efficiency specifically in functional areas such as grade encoding, and academic record verification. The instrument is divided into sections to capture the specific requirements of the Registrar's office and faculty, focusing on how automation reduces administrative burdens.

In addition, a process observation checklist is used during enrollment and record retrieval activities to document actual procedures, turnaround times, and the frequency of manual errors. This instrument enables an objective comparison between manual workflows and the proposed automated state, revealing gaps that the ASRMS must address. A document review form complements these tools, allowing for the examination of existing student files, request forms, and legacy databases to determine current data digitization and storage practices.

To ensure accuracy, the instruments undergo content validation by three experts in information technology and educational management. A pilot test will also be conducted to measure reliability using Cronbach's alpha, ensuring internal consistency. Collectively, these instruments provide the empirical foundation for analyzing how automation directly improves operational efficiency and for designing a system tailored specifically to TMCC's institutional context.

3.5. Statistical Treatment of Data

The statistical treatment of data for this project involves using descriptive and inferential statistics to evaluate how the Automated Student Record Management System (ASRMS) impacts TMCC's

operational efficiency. Frequency, percentage, mean, and ranking will be used to summarize the demographic profile of the respondents and describe their assessment of the current manual record-keeping systems. This descriptive analysis will pinpoint the specific bottlenecks in student record handling and the baseline level of efficiency in the Registrar's office.

The weighted mean will be applied to interpret the level of user acceptance and system performance based on key criteria: processing efficiency, data accuracy, accessibility, and reliability. To determine the effectiveness of the automation, the study will analyze the difference in "Turnaround Time" and error rates before and after the system's implementation. Furthermore, the Pearson correlation coefficient will be employed to determine the relationship between the system's technical usability and the resulting improvement in administrative operational speed.

Data will be processed using statistical software such as SPSS to ensure accuracy and objectivity. The results will serve as the primary evidence for implementing the system's effectiveness and for refining its functions to meet the specific needs of TMCC's academic records lifecycle. This quantitative approach ensures that the conclusions regarding operational efficiency are supported by empirical evidence, enabling an objective assessment of the system's impact on the institution.

3.6. System Analysis

The Automated Student Records Management System at Trece Martires City College serves multiple stakeholders involved in maintaining and accessing student academic information. Students use the system to securely view their academic records and submit updates to their personal information, with changes applied only after approval by the registrar.

Administrative staff, particularly in the Registrar's Office, use the system to create, read,

update, and delete student records, thereby improving accuracy and reducing manual workload. Meanwhile, administrators manage system settings and utilize system-generated reports to monitor academic performance, evaluate operational efficiency, and ensure compliance with institutional policies.

3.6.1. Business Process Modeling and Notation (BPMN)

Business Process Modeling and Notation (BPMN) serves as a structured model of the admission and records management process by visualizing workflows, decision gateways, and data movement between administrative and automated systems. Gateways function as decision points that determine application progression or record update handling, while data objects and data stores represent the lifecycle of records—transitioning from physical storage.

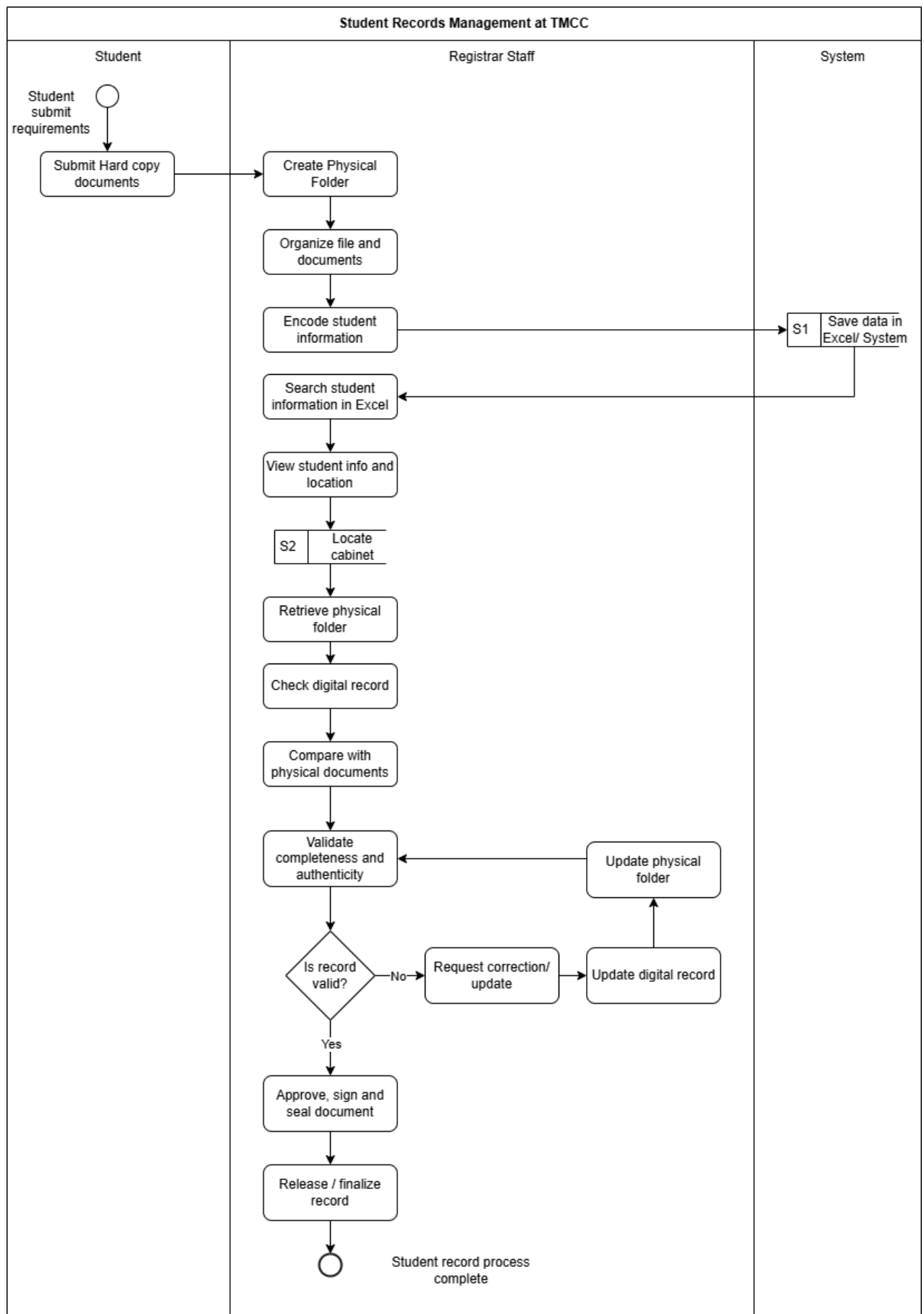


Figure 2.1 AS-IS for Current Student Records Management Process

The figure illustrates the Business Process Model and Notation (BPMN) of the current student records management system, integrating the processes of storing, retrieving, and verifying student records into a single continuous workflow. The process begins with a single start event, where the student submits requirements or initiates a record update handling.

The diagram is divided into swimlanes representing the key participants, namely the Student, Registrar Staff, and Records Storage/System. In the storing phase, the registrar staff creates a physical student folder, organizes the submitted documents, encodes the student's information into a digital system such as an Excel file, and stores the physical records in a filing cabinet. This reflects the dual-record approach where physical files serve as the official records, while digital files act as an index for faster searching.

The process then transitions into the retrieval phase, where the registrar staff searches for student information using the digital system, identifies the storage location, and retrieves the corresponding physical folder from the cabinet. This highlights the semi-manual nature of the system, as digital tools assist in locating records, but physical retrieval is still required.

In the verification phase, the registrar staff checks the digital record and compares it with the physical documents to validate accuracy, completeness, and authenticity. A decision gateway determines whether the record is valid. If valid, the document is approved, signed, and sealed. If not, a correction and updating process is performed, requiring modifications to both digital and physical records before re-validation.

The process concludes with a single end event, indicating the completion of the student

record transaction. Overall, the diagram emphasizes the reliance on both physical and digital systems, as well as the presence of manual intervention, redundancy, and potential inefficiencies within the current semi-manual process.

The Automated Student Records Management System (ASRMS) is designed to centralize and streamline the management of student records at TMCC through a secure and structured digital system. It focuses on role-based access for Students, Registrar Staff, and Administrators, supporting efficient CRUD operations for student information and academic records. The system allows students to view their records, while registrar staff are responsible for creating, updating, and deleting record data, ensuring accuracy and consistency. Administrators oversee user management and monitor system activities through transaction logs, promoting accountability and data integrity. Overall, ASRMS aims to reduce manual processing errors and improve the efficiency and reliability of student records management.

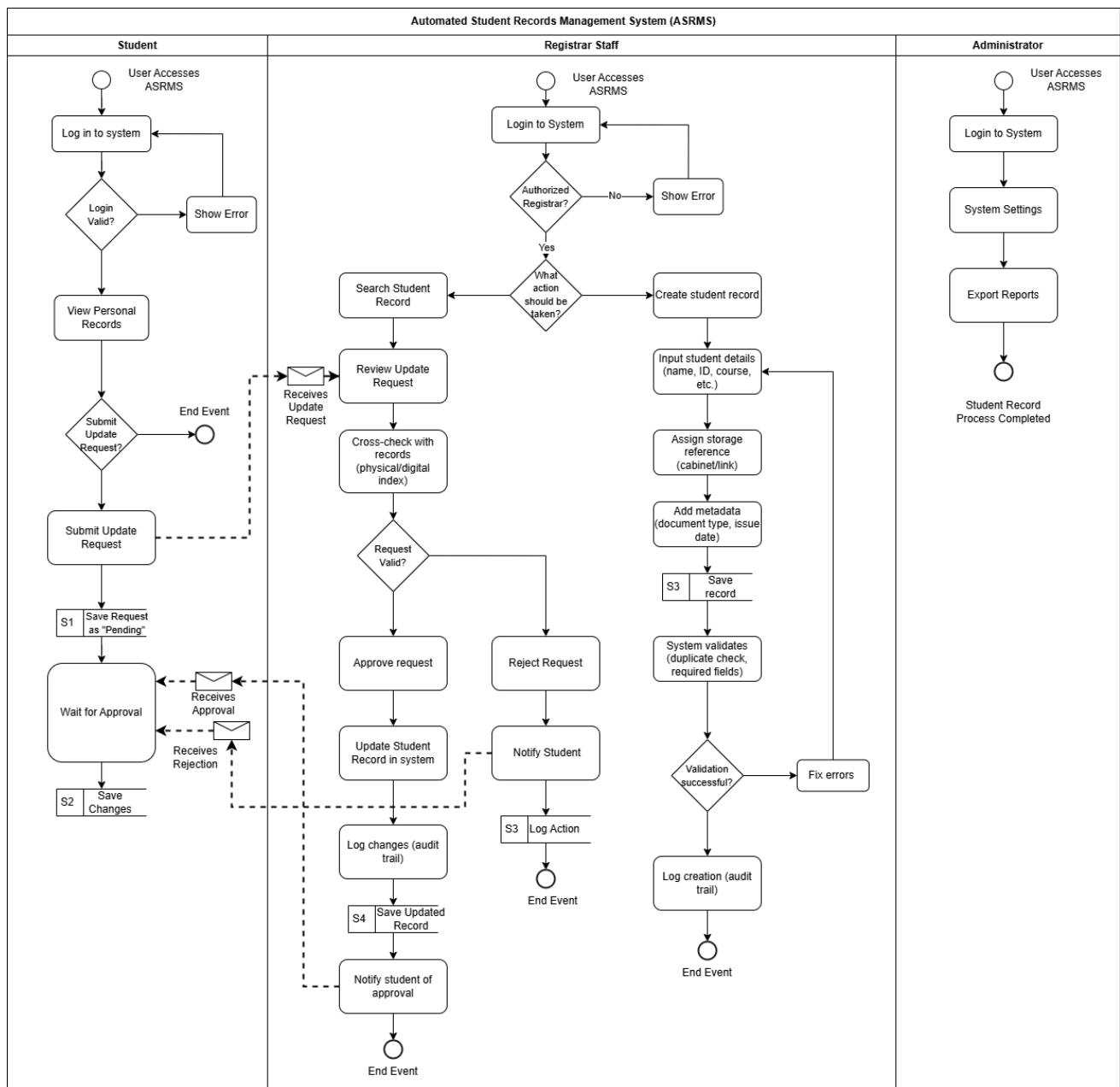


Figure 2.2 TO-BE BPMN of Proposed Automated Student Records Management System

The figure represents the Business Process Model and Notation (BPMN) of the student records management system. It illustrates the workflow of an Automated Student Records Management System (ASRMS), involving three primary actors: the student, the registrar staff, and the administrator. The process begins with the student logging into the system. If the login credentials are invalid, an error message is displayed; otherwise, the student gains access to view their personal records. From there, the student can submit a request to update their information and optionally upload

supporting documents. Once submitted, the request is saved as pending, and the student waits for the registrar's decision. If the request is approved, the student is notified and the changes are saved; if rejected, the student is informed accordingly.

On the registrar staff side, the process also starts with system login and authentication. Once authorized, the registrar can either search for an existing student record or create a new one. When handling update requests, the registrar receives and reviews the submission, cross-checking it with existing physical or digital records. A decision is then made based on the validity of the request. If approved, the registrar updates the student record, logs the changes for audit purposes, saves the updated record, and notifies the student of the approval. If the request is rejected, the registrar logs the action and sends a notification to the student regarding the rejection. Additionally, when creating a new student record, the registrar inputs necessary details such as name, ID, and course, assigns a storage reference, adds relevant metadata, and performs system validation before saving. Any errors encountered must be corrected before finalizing, after which the creation is logged and the process ends.

Meanwhile, the administrator interacts with the system at a supervisory level. After logging in, the administrator monitors system operations, configures system settings, manages user access and permissions, and generates reports. Once administrative tasks are completed, the overall process concludes. The entire workflow ensures proper validation, accountability through audit trails, and efficient management of student records within the system.

3.6.2.Current System Analysis

TMCC currently operates with partially automated systems combined with largely manual processes in managing student records. Student academic records are often fragmented across departments, making retrieval, verification, and updating time-consuming and prone to errors. Administrative staff handle record updates manually, resulting in delays, data inconsistencies, and increased workload due to repetitive paperwork. These operational challenges highlight the need for a fully automated student records management system to centralize data, streamline record processing, and improve overall operational efficiency.

3.6.3. Data Gathering Procedure

The data gathering procedure for the Automated Student Record Management System (ASRMS) for Trece Martires City College will follow a structured and evidence-based process to ensure that all information collected accurately reflects the institution's operational realities. The study utilizes a descriptive-developmental research design with a quantitative focus. During the planning stage, preliminary consultations will be held with TMCC administrators, the Registrar, and IT personnel to identify core academic record-keeping processes that require automation. This step will establish a baseline understanding of existing manual workflows in student registration, grade archiving, and document retrieval.

For data collection, the researcher will employ multiple instruments, including structured questionnaires and process observation checklists. Questionnaires will be used to assess the current level of automation and operational efficiency within the Registrar's Office and Admissions Department. Process observations will be carried out during grade encoding and transcript processing to record specific inefficiencies, such as data redundancies and manual verification delays. Document analysis will also be performed to review existing Student Record Management protocols and

historical ICT usage data. The study will also examine the procedures performed by authorized registrar staff in manually encoding, validating, updating, and approving student academic records using registrar account interfaces within the proposed system.

This evidence-based approach ensures that the development of the ASRMS is grounded in the actual needs of TMCC's staff and students. By measuring current turnaround times and error rates during the observation phase, the study will have the necessary data to evaluate the system's eventual effectiveness in improving the college's overall operational efficiency.

3.6.4. Data Flow Diagram

The **Data Flow Diagram (DFD)** for the Automated Student Records Management System (ASRMS) provides a visual representation of how data moves through the system, from input to processing and output. It illustrates the interactions between students, the registrar, and administrators, highlighting how student records are viewed, updated, verified, and managed. The DFD identifies the flow of information between processes, data stores, and external entities, ensuring that academic and personal data are handled accurately and securely. By mapping these data interactions, the diagram helps stakeholders understand the system's operations, streamline workflows, and enhance the efficiency and reliability of student record management.

3.6.5. Context Diagram

The Context Diagram of the Automated Student Records Management System (ASRMS) provides a high-level overview of the system as a single central process that interacts with its external entities, namely the Student, Registrar, and Administrator. It defines the system boundary and shows how data flows between these users and the ASRMS without detailing internal processes. The diagram

highlights how students access and request their academic and personal records, how registrars manage, verify, and update student information, and how administrators oversee the system by managing users and generating reports. Overall, it illustrates how ASRMS serves as a centralized platform for efficient, secure, and organized student records management.

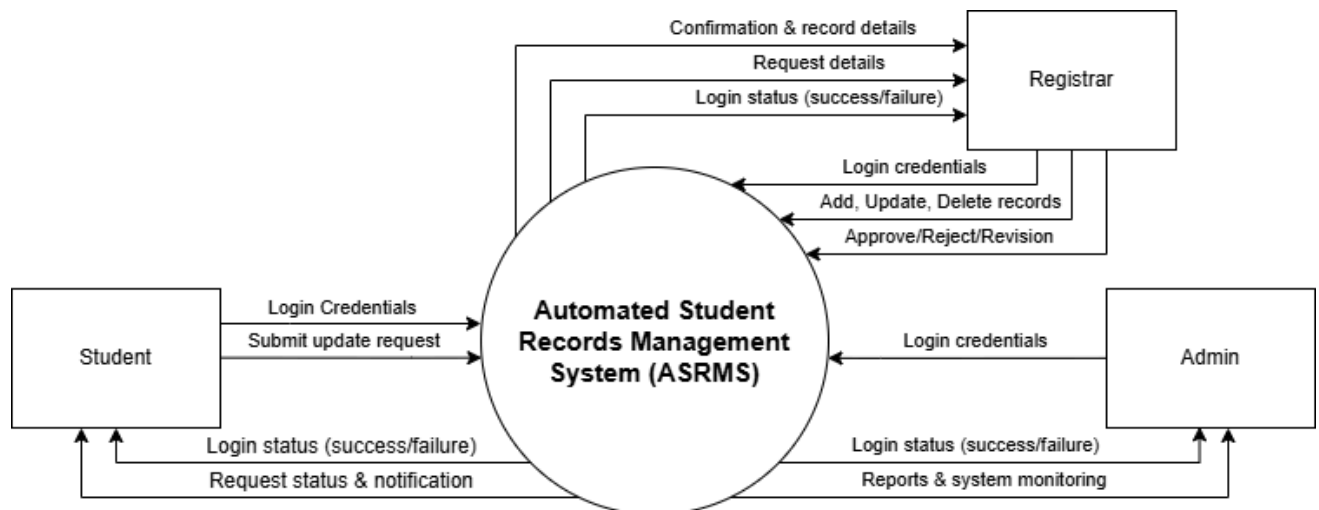


Figure 3 Level 0 Context Diagram of Automated Student Records Management System

The Level 0 Context Diagram illustrates the overall interaction between external users and the Automated Student Records Management System (ASRMS). At the center of the diagram is the ASRMS, which serves as the main system that processes and manages student record update handling. It connects to three primary external entities: the Student, the Registrar, and the Admin, each with specific roles and interactions.

The process begins with the Student, who provides login credentials to access the system and submits requests for record updates. In return, the system sends back login status (whether successful or failed) as well as updates on the request status and notifications. This shows that the student

primarily initiates requests and monitors their progress through the system.

The Registrar interacts more extensively with the system. After logging in, the registrar can view request details, confirm records, and perform actions such as adding, updating, or deleting student records. The registrar is also responsible for reviewing submitted requests and deciding whether to approve, reject, or request revisions. The system provides feedback to the registrar through login status and displays necessary request and confirmation details.

On the other hand, the Admin is responsible for higher-level system control and monitoring. The admin logs into the system to oversee operations and access reports. The ASRMS provides the admin with system monitoring data and reports, along with login status updates, ensuring that the system runs efficiently and securely.

Overall, the diagram presents a high-level view of how data flows between users and the system, emphasizing the roles of each actor and how the ASRMS acts as the central hub for authentication, request processing, record management, and system monitoring.

3.6.6. Level 1 Diagram

The Level 1 Data Flow Diagram (DFD) of the Automated Student Records Management System (ASRMS) provides a more detailed view of how the system operates internally. It expands the single process shown in the Level 0 diagram into several interconnected subprocesses, illustrating how data flows within the system. This diagram focuses on the movement and transformation of data as it passes through different functional components, offering a clearer understanding of how the system handles user interactions, processes requests, and manages student records efficiently.

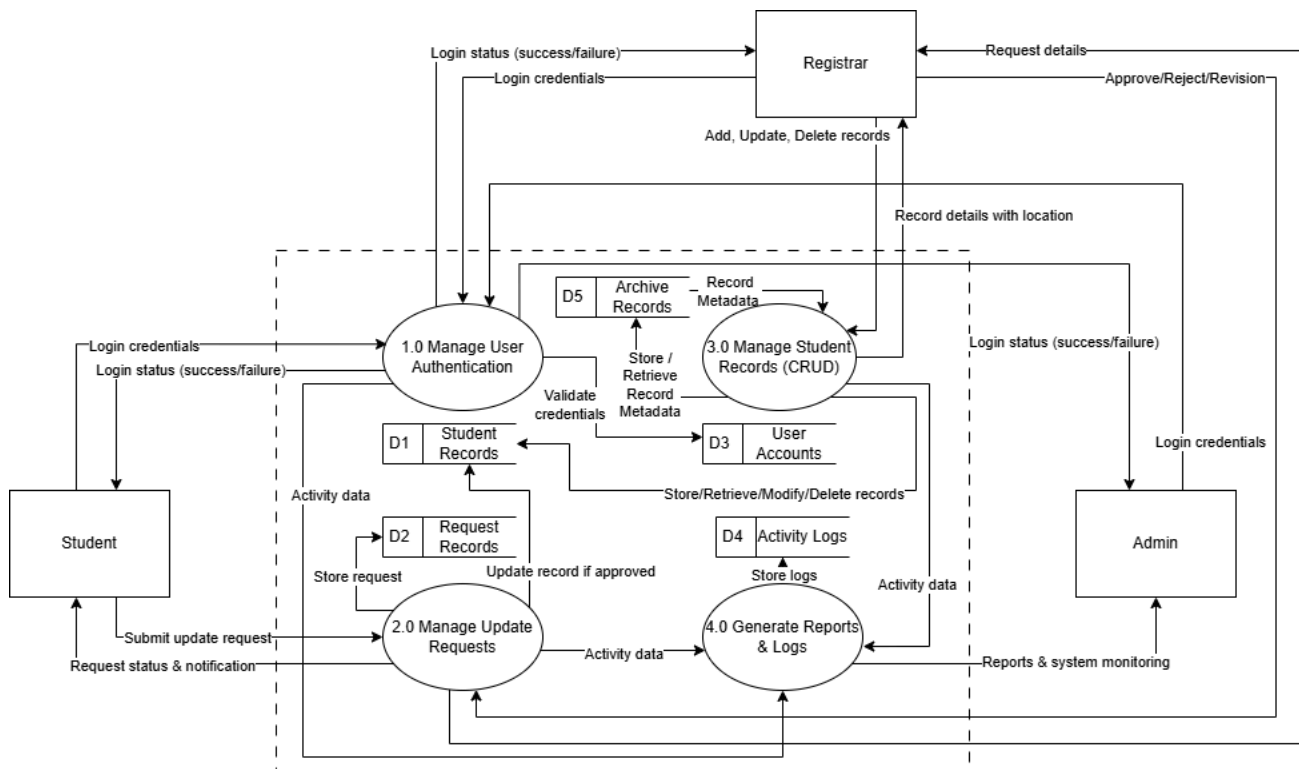


Figure 4 Level 1 Data Flow Diagram of Automated Student Records Management System

The Level 1 Data Flow Diagram (DFD) illustrates the internal processes and data interactions of the proposed student record management system. It provides a detailed breakdown of how data moves between external entities, namely the Student, Registrar, and Admin, and the core system processes, which include user authentication, request management, student record management, and report generation.

The system begins with the authentication process, where both students and administrators submit their login credentials through the Manage User Authentication module. These credentials are validated against stored user account data to determine whether access should be granted or denied. Upon successful authentication, users are allowed to proceed to their respective functionalities within the system.

Once logged in, students interact with the system primarily by submitting update requests through the Manage Update Requests process. These requests are stored in the request records data store and may include changes or corrections to their personal or academic information. The system tracks the status of each request and provides notifications back to the student regarding approval, rejection, or required revisions.

The management of student records is handled by the Manage Student Records (CRUD) module. This process allows authorized registrar staff to create, read, update, and delete student records through manual data encoding using registrar staff account interfaces. Registrar personnel are responsible for validating, approving, and maintaining the accuracy of academic and personal information before changes are officially applied to the system. The process retrieves and updates data from the student records data store and ensures that only validated and approved changes are reflected within the system. Students are limited to viewing approved academic records and submitting requests for personal information updates, which remain subject to registrar review and approval.

Finally, the Generate Reports and Logs process compiles data from activity logs and other system records to produce comprehensive reports. These reports are primarily accessed by the administrator for system monitoring, performance evaluation, and decision-making purposes. Overall, this Level 1 DFD demonstrates a structured and secure flow of information within the system, highlighting how data is processed, stored, and utilized across different modules. It emphasizes the integration of authentication, record management, request handling, and reporting functionalities to support efficient and reliable system operations.

3.6.7.Expllosion Levels

The Explosion Level Data Flow Diagram of the Automated Student Records Management System (ASRMS) provides an even more detailed view of a specific process from the Level 1 DFD by breaking it down into smaller, more precise subprocesses. This level focuses on the internal workings of a single major function, showing step-by-step how data is processed, validated, and transformed within that process. This level serves to refine and elaborate on the system's functionality, making it easier to analyze, design, and implement each component of the ASRMS.

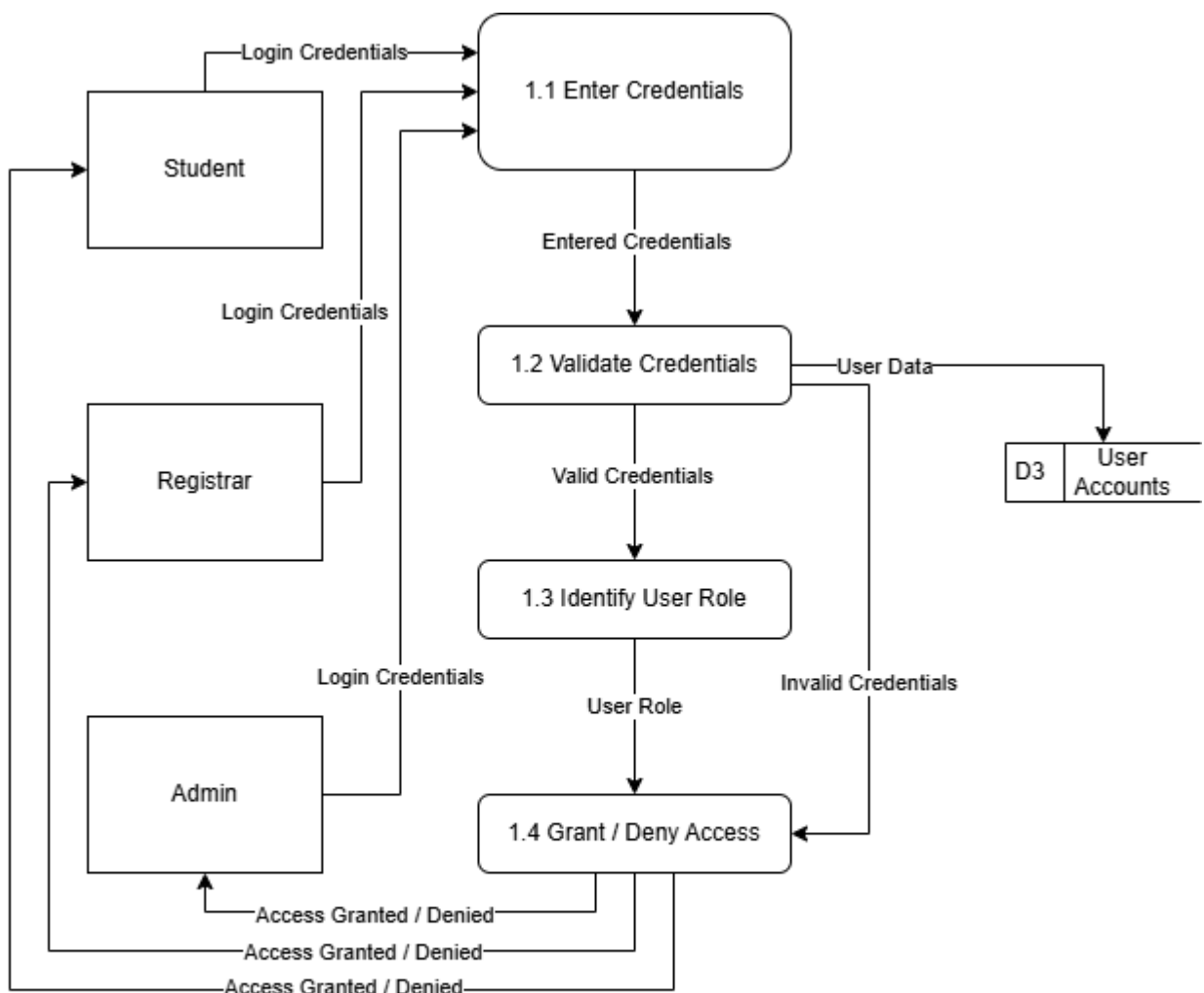


Figure 5.1 Manage User Authentication Explosion Level

The Level 2 Explosion level for the Manage User Authentication process provides a more

detailed view of how user authentication is handled within the system. It expands the Level 1 authentication module into a series of interconnected sub-processes that ensure secure and controlled access to the system. The authentication process begins when external entities—namely the Student, Registrar, or Admin—submit their login credentials. These credentials are received by the Enter Credentials process, where user input such as username and password is captured and prepared for validation. Once entered, the credentials are forwarded to the Validate Credentials process. In this stage, the system compares the entered credentials with stored user data retrieved from the User Accounts data store. This validation step ensures that only registered and legitimate users are allowed to proceed further. If the credentials do not match any records in the system, the process identifies them as invalid, and access is denied.

If the credentials are successfully validated, the system proceeds to the Identify User Role process. Here, the system determines the role associated with the authenticated user—whether they are a student, registrar, or administrator. This role identification is critical, as it defines the level of access and the permissions granted to the user within the system.

Following role identification, the Grant or Deny Access process makes the final decision regarding system access. If all authentication conditions are satisfied, access is granted, and the user is allowed to interact with the system according to their role-based permissions. Conversely, if any step fails—such as invalid credentials—the system denies access and may prompt the user to retry. Overall, this Level 2 DFD illustrates a secure and systematic approach to user authentication. It emphasizes the importance of credential validation, role identification, and controlled access, ensuring that only authorized users can interact with sensitive system functionalities and data.

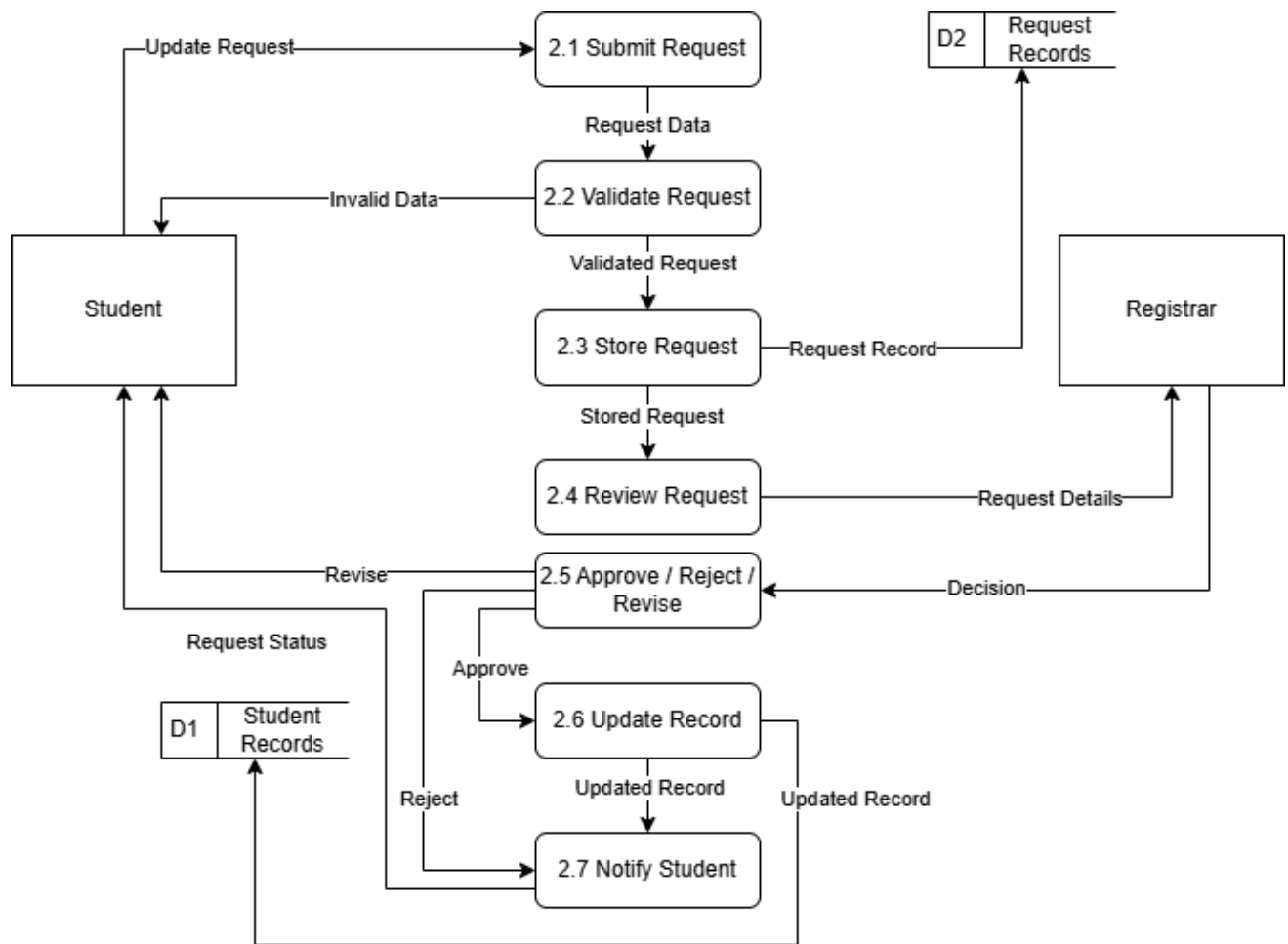


Figure 5.2 Manage Update Request Explosion Level

The Level 2 Explosion Level of Manage Update Request process illustrates the detailed workflow involved in handling student-initiated requests for updating academic or personal records within an institutional system. This process ensures that all requests are properly submitted, validated, reviewed, and either approved or rejected while maintaining data integrity and proper communication between stakeholders.

The process begins when the Student submits an update request through Process 2.1: Submit Request. The submitted data is then passed to Process 2.2: Validate Request, where the system checks the completeness and correctness of the provided information. If the data is found invalid, it is returned to the student for correction. Otherwise, a validated request proceeds to the next stage. In

Process 2.3: Store Request, the validated request is recorded and stored in the Request Records (D2) data store. This ensures that all requests are logged for tracking and auditing purposes. The stored request is then forwarded to Process 2.4: Review Request, where the Registrar examines the request details.

Following the review, the system moves to Process 2.5: Approve / Reject / Revise, where the registrar makes a decision. If revisions are required, the request is sent back to the student for modification. If approved, the process continues to Process 2.6: Update Record, where the necessary changes are applied to the Student Records (D1) data store. If rejected, the request does not proceed to record updating.

Finally, in Process 2.7: Notify Student, the system communicates the outcome of the request (approved, rejected, or requiring revision) back to the student. This ensures transparency and keeps the student informed about the status of their request. Overall, this process enforces a structured and controlled approach to managing update requests by incorporating validation, decision-making, record management, and user notification. It enhances data accuracy, accountability, and efficiency within the institution's record management system.

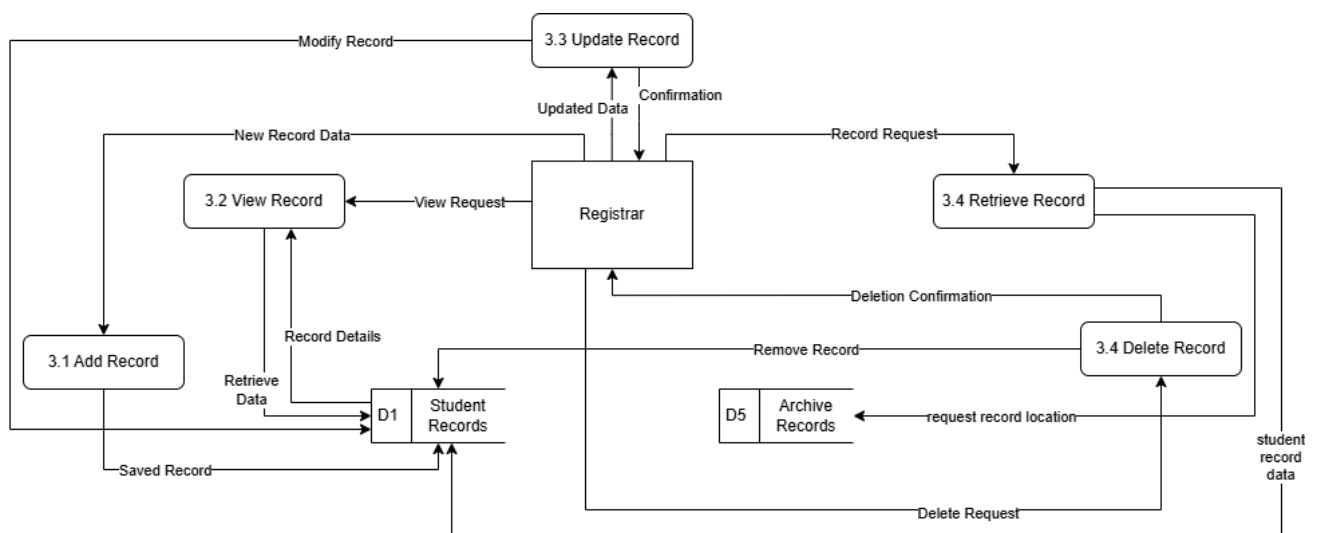


Figure 5.3 Manage Student Records Explosion Level

The Level 2 Explosion level of Manage Student Records (CRUD) process illustrates how the system handles the creation, retrieval, updating, and deletion of student records under the supervision of the registrar. The process begins when the registrar inputs new record data, which is handled by 3.1 Add Record, where the information is stored and saved into the student records data store (D1). For accessing existing information, the registrar initiates a request through 3.2 View Record, allowing retrieval of stored student data for review or verification purposes.

When modifications are necessary, the process moves to 3.3 Update Record, where updated data is applied and reflected in the data store to ensure that records remain accurate and current. In cases where a record needs to be removed, the registrar performs 3.4 Delete Record, which deletes the selected data and updates the system accordingly, followed by a deletion confirmation. Throughout the process, the student records data store (D1) serves as the central repository, supporting all CRUD operations by storing, retrieving, updating, and removing data as required. This structured workflow ensures efficient, consistent, and secure management of student records within the system.

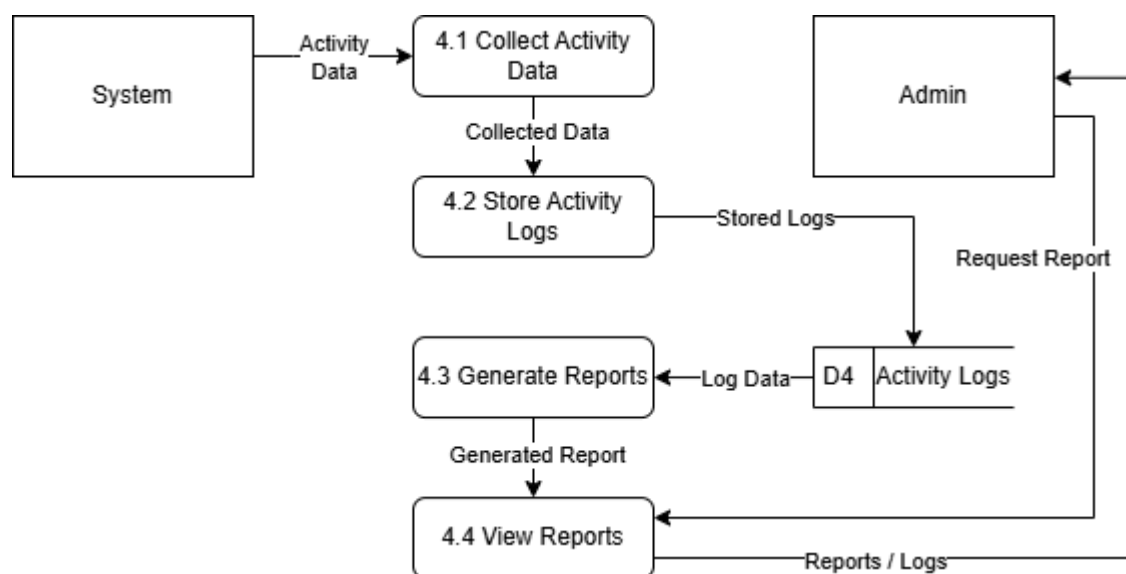


Figure 5.4 Manage Logs and Reports Explosion Level

The Level 2 Explosion level of Manage Logs and Reports process provides a detailed decomposition of the system's internal functionalities related to activity tracking, log storage, and report generation. This level elaborates on how data flows between system components and external entities, specifically the System and the Admin.

The process begins with 4.1 Collect Activity Data, where the system captures activity data generated from various system operations. This data may include user actions, system events, and transaction records. The primary purpose of this subprocess is to ensure that all relevant operational activities are accurately recorded for monitoring and auditing purposes. Following data collection, the process proceeds to 4.2 Store Activity Logs, where the collected data is systematically stored in a centralized data repository, identified as D4: Activity Logs. This data store acts as a persistent storage unit that maintains structured logs, enabling efficient retrieval and long-term record keeping. The storage mechanism ensures data integrity, consistency, and availability for future processing.

The next subprocess, 4.3 Generate Reports, retrieves log data from the Activity Logs data store. In this stage, the system processes and analyzes the stored logs to produce meaningful reports. These reports may include summaries of system usage, performance metrics, or audit trails. The transformation of raw log data into structured reports facilitates better decision-making and system evaluation.

Finally, 4.4 View Reports enables the Admin to access and review the generated reports. The Admin initiates this process by sending a request for reports, which triggers the system to present the relevant logs and generated outputs. This subprocess ensures that authorized users can interact with the reporting module, interpret system activities, and derive insights. Overall, this Level 2 diagram illustrates a structured workflow where activity data is collected, stored, processed, and presented. It

highlights the interaction between system processes and the Admin, emphasizing the importance of efficient log management and reporting in maintaining system transparency, accountability, and performance monitoring.

3.7. Program Design

The program design of the Automated Student Records Management System (ASRMS) defines the overall structure, components, and operational environment of the system. It outlines how the software and hardware components interact to ensure efficient, secure, and reliable management of student records. The design is divided into external and internal components to clearly distinguish between the system's operational environment and its internal architecture. This structured approach ensures that the system meets institutional requirements for data accuracy, security, and scalability while supporting the daily operations of the registrar's office.

3.7.1. External Design

The Student Records Management System will operate on a dedicated server located within the registrar's office to ensure centralized and secure data management. The system is designed to function within a local area network (LAN) environment and will remain isolated from the public internet to protect sensitive student information from external cyber threats.

The institution will provide a dedicated server equipped with a high-performance multi-core processor, running on Windows Server 2025 Standard Edition, and supported by a minimum of 16 GB of RAM to efficiently handle multiple concurrent database operations. For storage, the system will utilize a high-capacity Solid State Drive (SSD) to host the operating system and database, ensuring fast

data access and retrieval. Additionally, a secondary Hard Disk Drive (HDD) will serve as a backup repository to maintain data redundancy and support recovery processes.

The registrar's workstation will connect to the server through a network switch using CAT6 Ethernet cables, ensuring a stable and high-speed connection. To further enhance system reliability, an Uninterruptible Power Supply (UPS) will be installed to prevent data loss and system damage during unexpected power interruptions.

3.7.2. Internal Design

The internal architecture of the Student Records Management System follows the Model-View-Controller (MVC) design pattern. This approach divides the system into three main components: the user interface (View), the business logic (Controller), and the data management layer (Model). This separation ensures modularity, making the system easier to maintain, update, and scale.

The system is developed using Laravel (PHP framework) for the backend and React JS for the frontend interface. Supporting technologies such as PHP and JavaScript are utilized to handle server-side processing and client-side interactions, respectively.

The View component represents the user interface, which is responsible for presenting data to users and capturing their inputs. It consists of dynamic web pages, forms, and dashboards developed using React JS, along with HTML, CSS, and JavaScript. This layer enables users such as students, registrars, and administrators to interact with the system efficiently by viewing records, submitting requests, and accessing reports in a responsive and user-friendly environment.

The Controller serves as the intermediary between the user interface and the database. Implemented through Laravel's controller structure, it processes user inputs, applies validation rules,

and manages system operations. For instance, when a registrar submits a request to update student records, the controller validates the request, ensures compliance with system rules, and communicates with the model to execute the necessary database operations.

The Model component handles data management and database interactions. Using Laravel's Eloquent ORM, it is responsible for executing operations such as inserting new student records, updating academic information, retrieving archived documents, and managing metadata, including cabinet locations for physical records. The model ensures data integrity by enforcing predefined structures, relationships, and constraints before storing information in the database.

3.8. Data Structure of the Program

The data structure for the Automated Student Records Management System (ASRMS) is designed using a relational database schema that focuses on efficient CRUD operations and accurate records management. At the core of the system is the User table, which implements Role-Based Access Control (RBAC) to distinguish between Students, Registrar Staff, and Administrators. Each user is assigned a unique user ID, ensuring secure login and controlled access to system functionalities. This structure guarantees that only authorized users can access or modify specific data within the system.

The Student table serves as the central repository of student personal information, where each student is identified by a unique student ID. This ID is used to link all related records, ensuring that data remains consistent and avoids duplication. Connected to this is the Student Records table, which contains the primary academic and record-related information managed in the system. This includes basic academic data and other relevant student record details. Registrar staff are responsible for performing CRUD operations—creating, reading, updating, and deleting records—while

students are limited to viewing their information only.

To maintain accountability and ensure data integrity, the system includes a Transaction Log table that records all actions performed by registrar staff and administrators. Each entry captures the user ID, the affected record ID, the type of operation (create, update, delete), and the timestamp of the action. This audit trail helps monitor changes, prevent unauthorized modifications, and support system transparency. This structured data design supports a centralized and organized approach to student records management, reducing redundancy and minimizing human errors commonly found in manual or semi-manual systems. By focusing on essential CRUD functionalities and secure data handling, the ASRMS ensures reliable, efficient, and accurate management of student records.

3.8.1. Use Case Diagram

The Use Case Diagram for the Automated Student Records Management System (ASRMS) illustrates the functional boundaries of the system and the interactions between the primary actors: Students, Registrar Staff, and Administrators. This diagram defines the scope of the software by showing how different user roles consume the specific features of the request lifecycle.

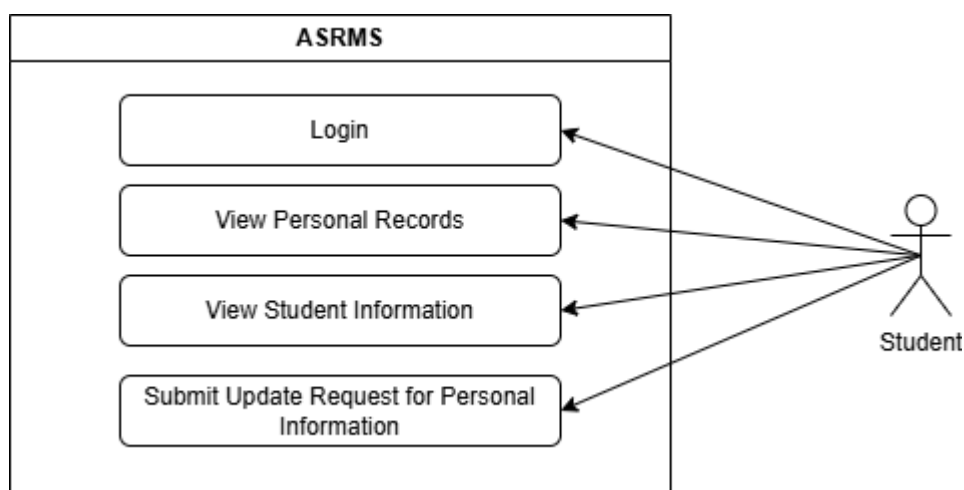


Figure 6.1 Student Use Case

The figure illustrates the interaction of the **Student** actor with the Automated Student Records Management System (ASRMS). The Student can log into the system to access and view personal records and student information. Additionally, the Student is allowed to submit update requests for personal information, which are subject to approval by the registrar staff. These interactions provide a secure and user-friendly interface for students to access their records and request necessary updates while maintaining data integrity through controlled approval processes.

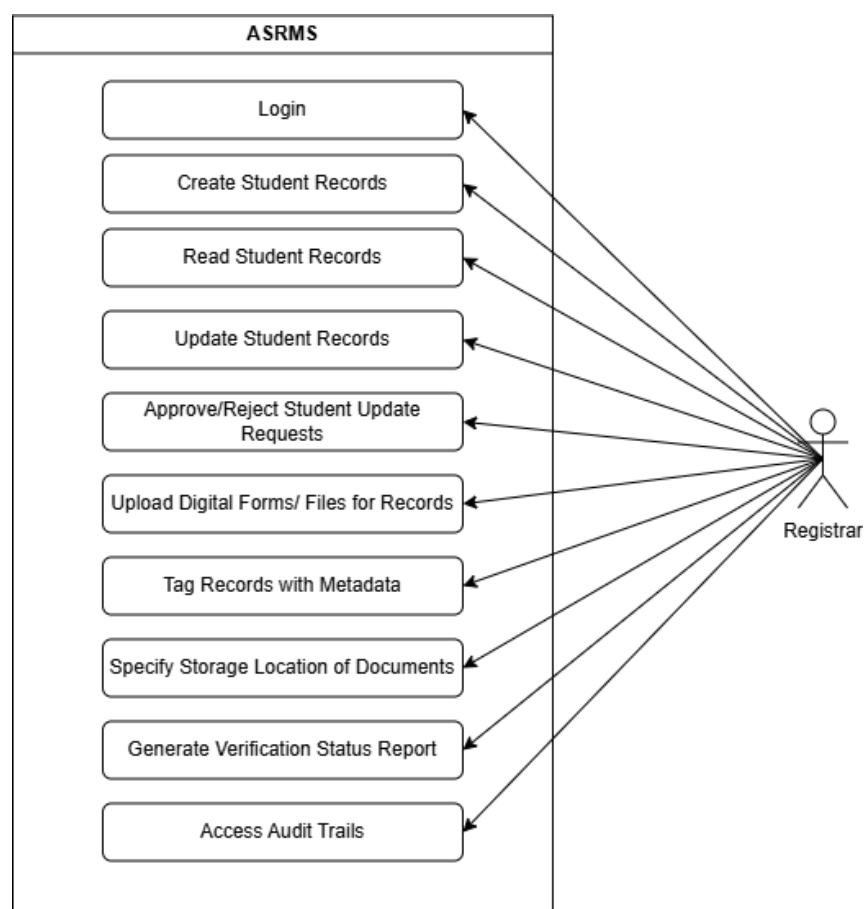


Figure 6.2 Registrar Staff Use Case

The figure illustrates the interaction of the **Registrar** Staff actor with the Automated Student Records Management System (ASRMS). The Registrar Staff is responsible for managing student records through various system functionalities, including logging into the system, creating, reading, updating, and deleting student records. Additionally, the staff handles the approval or

rejection of student update requests and uploads digital forms or files associated with records. The system also allows the Registrar Staff to organize records by tagging metadata and specifying storage locations for legal documents. Furthermore, the staff can generate verification status reports, and access audit trails for monitoring and security purposes. This comprehensive set of interactions ensures efficient, secure, and accountable management of student records within the system.

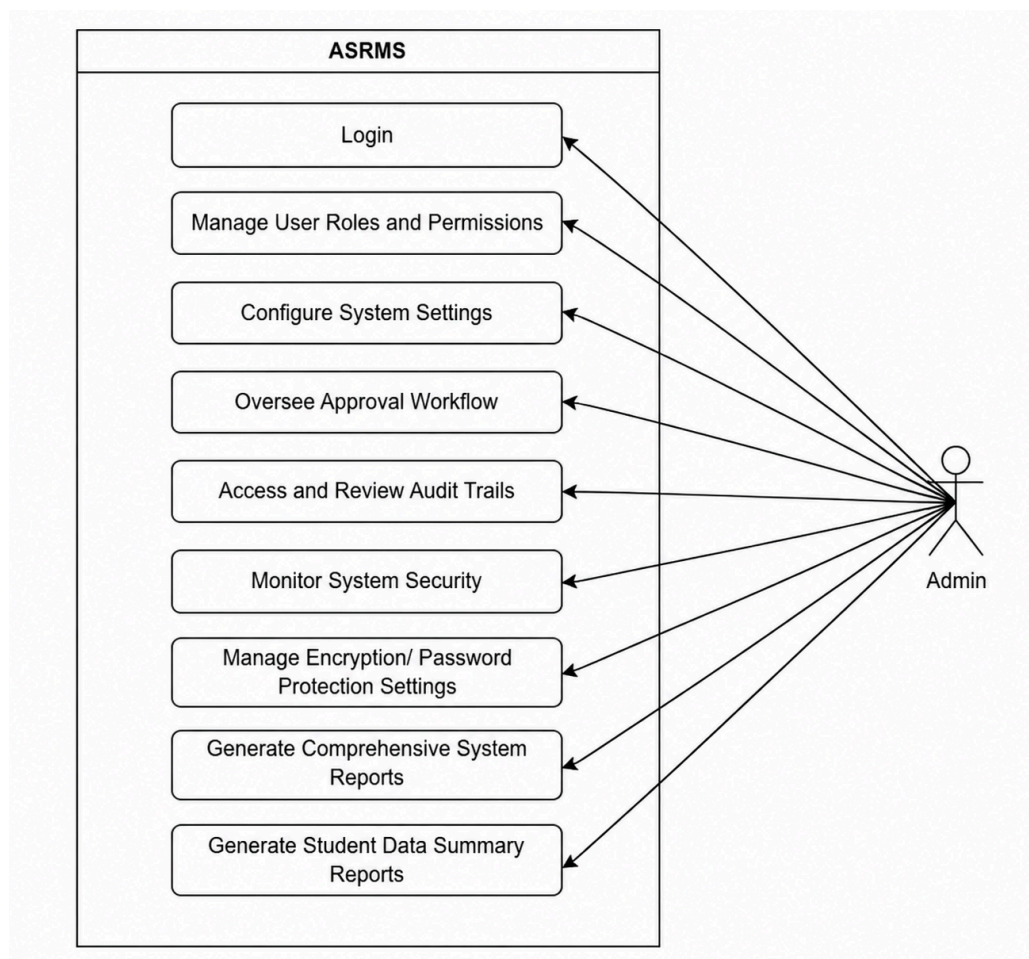


Figure 6.3 Admin Use Case

Finally, the figure illustrates the interaction of the Admin actor with the Automated Student Records Management System (ASRMS). The Admin is responsible for overseeing and monitoring the overall system operations, including logging in, managing user roles and permissions, and configuring system settings. The Admin also supervises approval workflows,

ensures proper system access control, and monitors system activity to maintain the security and integrity of student records.

In addition, the system enables the Admin to review audit trails for transparency and accountability. Furthermore, the Admin monitors system security, manages encryption and password protection settings, and generates comprehensive system and student data summary reports. These interactions ensure effective system governance, security monitoring, access control, audit transparency, and administrative oversight of overall system operations.

3.8.2. Class Diagram

The Class Diagram of the Automated Student Records Management System (ASRMS) illustrates the system's structure by defining its main classes, attributes, methods, and relationships. It highlights the role-based interactions among Student, Registrar Staff, and Administrator, as well as the central use of the StudentRecord class for managing academic data and the TransactionLog for monitoring system activities. This diagram provides a clear blueprint of how the system supports secure and efficient student records management.

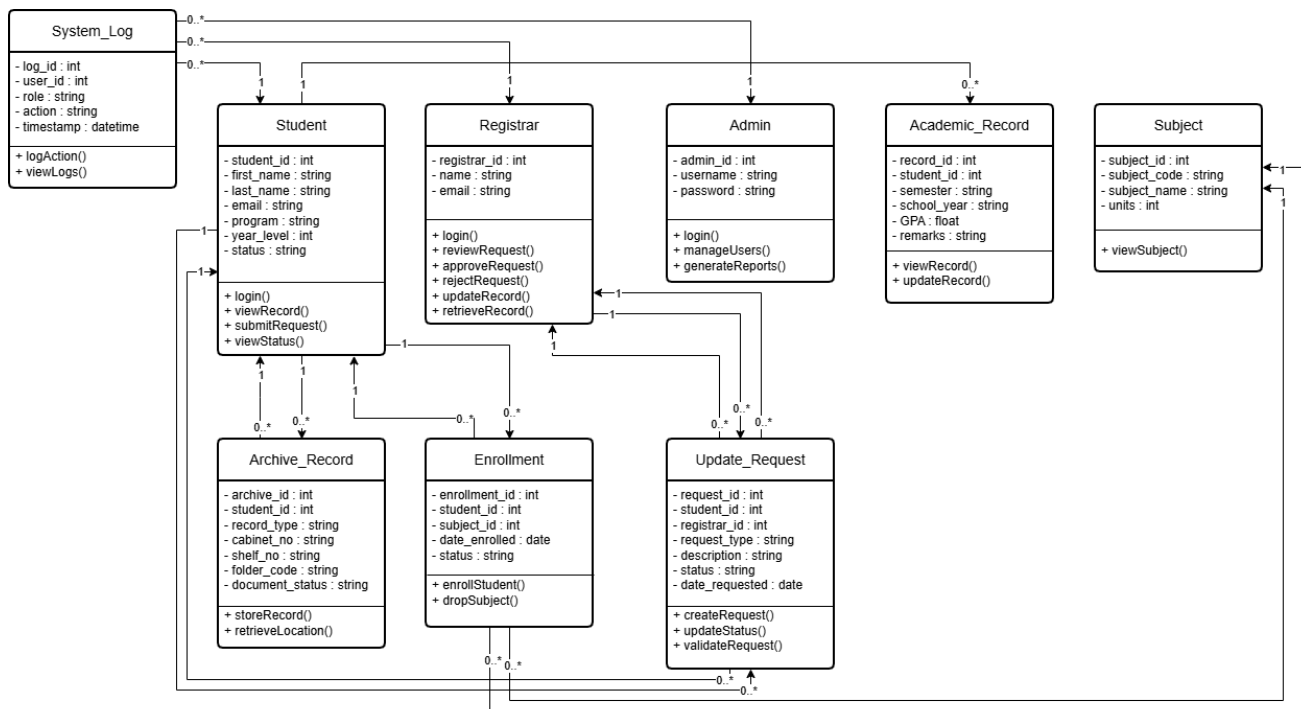


Figure 7 Class Diagram for Student Records Management System

The class diagram illustrates the object-oriented structure of the Automated Student Records Management System (ASRMS), defining the system's core classes, their attributes, methods, and relationships. At the center of the system is the Student class, which encapsulates student-related data such as student ID, name, email, program, year level, and status. It also provides key functionalities including login, viewing records, submitting requests, and tracking request status. A student is associated with multiple Academic_Record, Enrollment, and Archive_Record instances, reflecting the system's ability to manage academic performance, subject enrollment, and archived documents.

The Academic_Record class stores semester-based academic information, including school year, GPA, and remarks, with methods for viewing and updating records. Similarly, the Subject class defines course details such as subject code, subject name, and units, and allows users to view subject information. The Enrollment class acts as a linking class between students and subjects, supporting

operations such as enrolling in and dropping subjects, while maintaining attributes like enrollment date and status.

Request management is handled through the `Update_Request` class, which records student-initiated requests for updates or corrections. It includes attributes such as request type, description, status, and date requested, along with methods to create, validate, and update request statuses. These requests are processed by the `Registrar` class, which plays a key role in managing and approving student concerns. The registrar is equipped with functionalities such as reviewing, approving, rejecting requests, updating records, and retrieving student information.

Administrative control is represented by the `Admin` class, which is responsible for system management tasks such as user administration and report generation. Both admin and registrar activities are tracked by the `System_Log` class, which records user actions, roles, and timestamps, ensuring accountability and system transparency. Additionally, the `Archive_Record` class manages the storage and retrieval of archived student documents, including their physical or digital locations. Overall, the class diagram demonstrates a well-defined modular architecture where each class has specific responsibilities and interacts with others through clearly established relationships. This design supports scalability, maintainability, and efficient handling of student records, enrollment processes, request workflows, and system monitoring within the ASRMS.

3.8.3. Sequence Diagram

This section presents the sequence diagrams of the proposed Automated Student Records Management System (ASRMS), which depict the dynamic interactions among system users and components. These diagrams are utilized to model the step-by-step exchange of messages between the Student, Registrar, and Administrator, as well as the system and its underlying

database.

Each sequence diagram corresponds to a specific functional scenario of the system, such as viewing student records, submitting and processing update requests, managing student information through CRUD operations, and generating reports. The diagrams emphasize the chronological order of interactions, allowing for a clearer understanding of how processes are executed within the system.

Moreover, key system features such as data validation, role-based access control, approval mechanisms, and audit logging are reflected within the interaction flows. Conditional paths are represented using alternative (alt) frames to model decision-making processes, including approval, rejection, or revision of requests. Overall, these diagrams provide a comprehensive visualization of system behavior and serve as a foundation for system development and implementation.

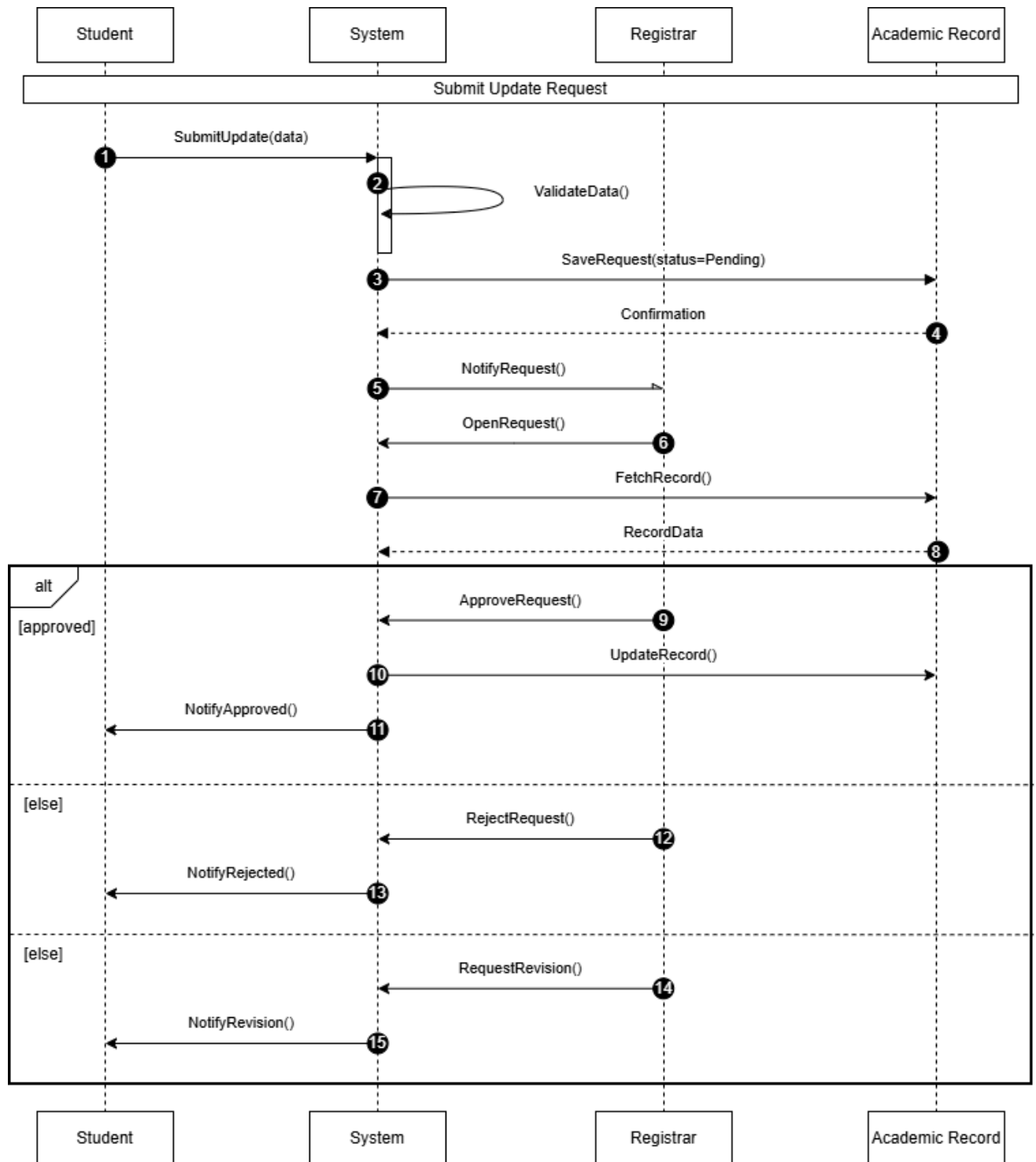


Figure 8.1 Sequence Diagram for Student (Submitting Update Request)

This sequence diagram presents the process of submitting and processing a student record update request in the Automated Student Records Management System (ASRMS). The

interaction begins when a student submits a request to update personal information through the system. The system performs initial validation checks to ensure that the submitted data meets the required format and completeness.

After validation, the request is stored with a “Pending” status and forwarded to the registrar for review. The registrar then evaluates the request by accessing existing records and verifying the accuracy of the submitted information. Based on the evaluation, the registrar may approve, reject, or request revisions.

If approved, the system updates the student’s record in the database and notifies the student of the successful update. If rejected or marked for revision, the system informs the student accordingly. This process ensures controlled data modification, maintaining the integrity and accuracy of student records through an approval-based workflow.

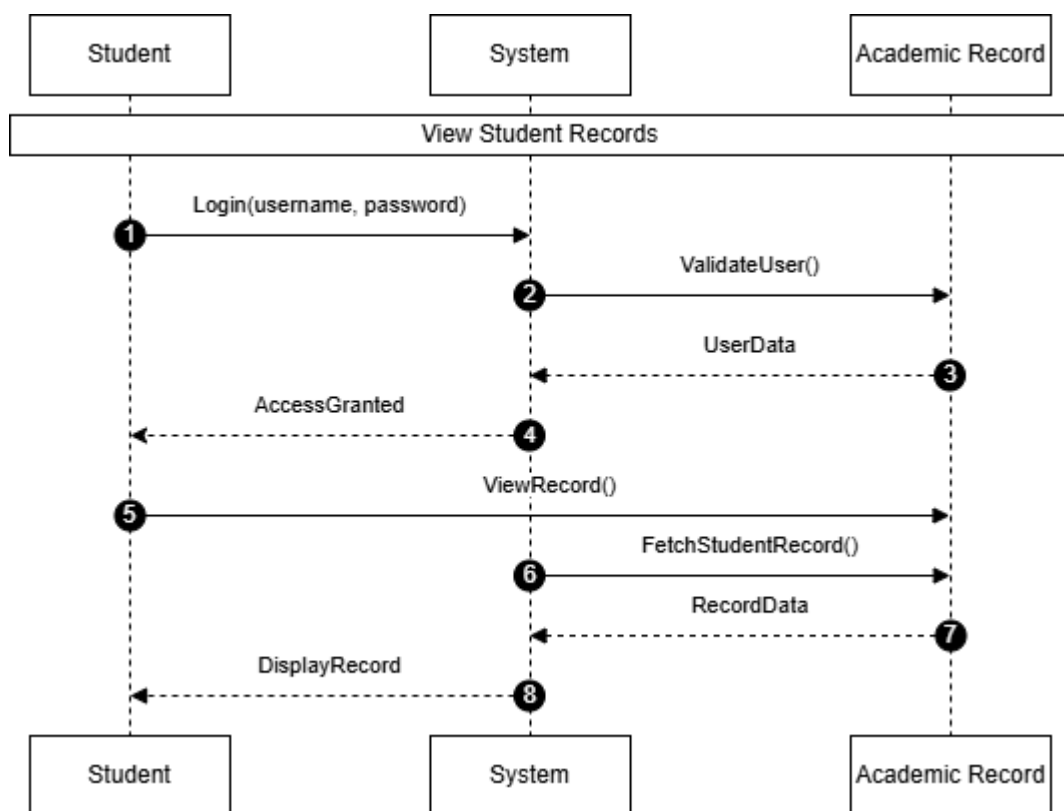


Figure 8.2 Sequence Diagram for Student (Viewing Student Records)

This sequence diagram illustrates the process of viewing student records in the Automated Student Records Management System (ASRMS). The interaction begins when the student logs into the system by providing valid credentials. The system then verifies the user information through the database and grants access upon successful authentication.

Once logged in, the student requests to view their personal record. The system retrieves the corresponding data from the database and displays it to the student. This process ensures that students can securely access their information in a read-only manner without directly modifying any stored data.

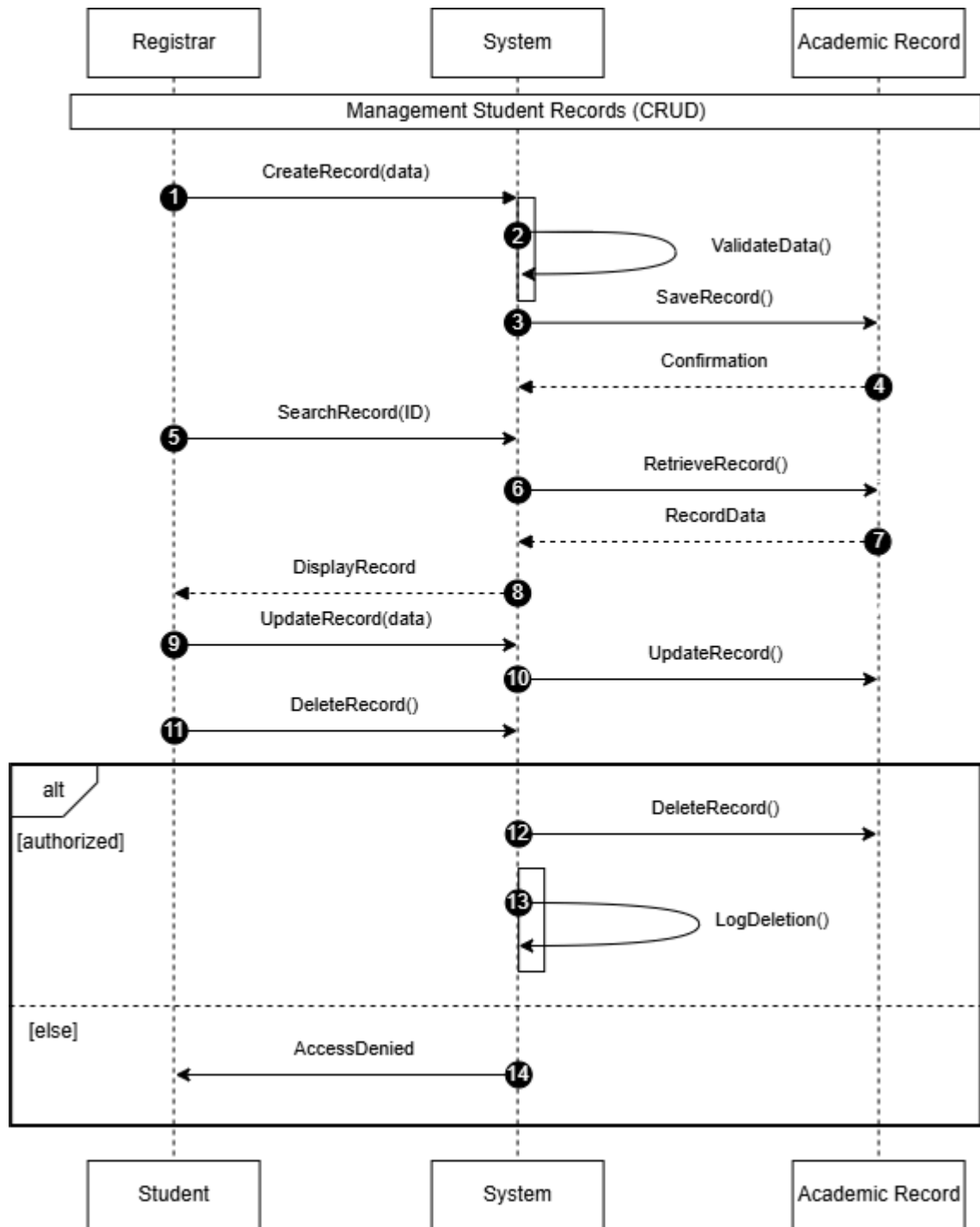


Figure 8.3 Sequence Diagram for Registrar (Managing Student Records)

This sequence diagram illustrates the process of managing student records through Create, Read, Update, and Delete (CRUD) operations in the Automated Student Records Management System (ASRMS). The process is primarily handled by the registrar, who is authorized to perform these operations.

The interaction begins with the creation of a new student record, where the system validates the input data before storing it in the database. The registrar can also retrieve and view existing records through search functionality. For updates, the registrar modifies the necessary information, which is then saved in the database.

In the case of record deletion, the system verifies whether the registrar has the appropriate authorization. If permitted, the record is deleted, and the action is logged in the system's audit trail. Otherwise, access is denied. This scenario ensures efficient and secure management of student data while maintaining accountability through validation and logging mechanisms.

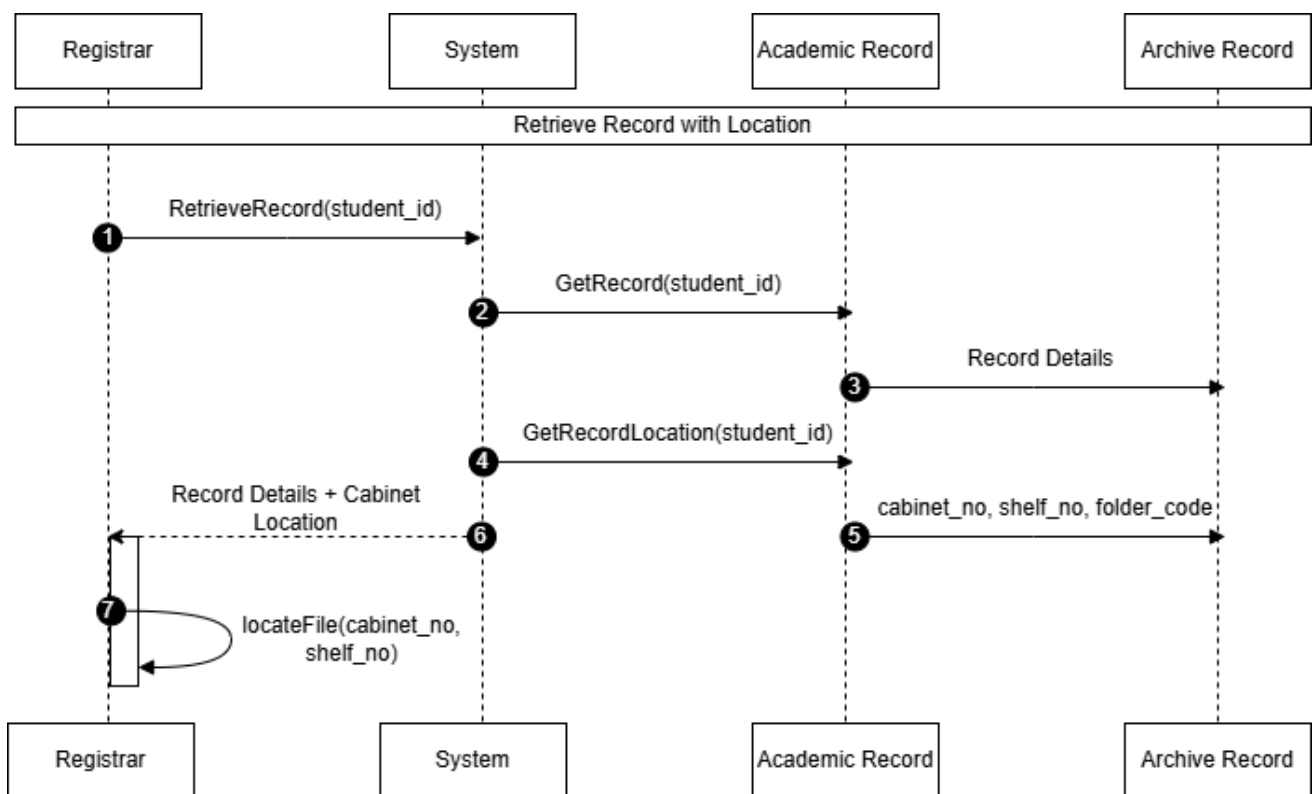


Figure 8.4 Sequence Diagram for Registrar (Retrieve Record with Location)

This sequence diagram illustrates the interaction between the Registrar, System, Academic Record, and Archive Record components in retrieving a student's academic record along with its corresponding physical storage location.

The process begins when the Registrar sends a `RetrieveRecord(student_id)` request to the System. Upon receiving the request, the System communicates with the Academic Record component to fetch the student's academic details through `GetRecord(student_id)`. The Academic Record then returns the requested record details to the System.

After retrieving the academic information, the System proceeds to obtain the physical file location by sending a `GetRecordLocation(student_id)` request to the Archive Record component. The Archive Record responds with the metadata, including the cabinet number, shelf number, and folder code, which indicate where the physical document is stored.

The System then consolidates both the academic record details and the cabinet location metadata and sends them back to the Registrar as `Record Details + Cabinet Location`. Finally, the Registrar performs a manual action to locate the physical file using the provided cabinet and shelf information. This diagram demonstrates how the system integrates both digital and physical record management by combining database retrieval with archive metadata to support efficient record access.

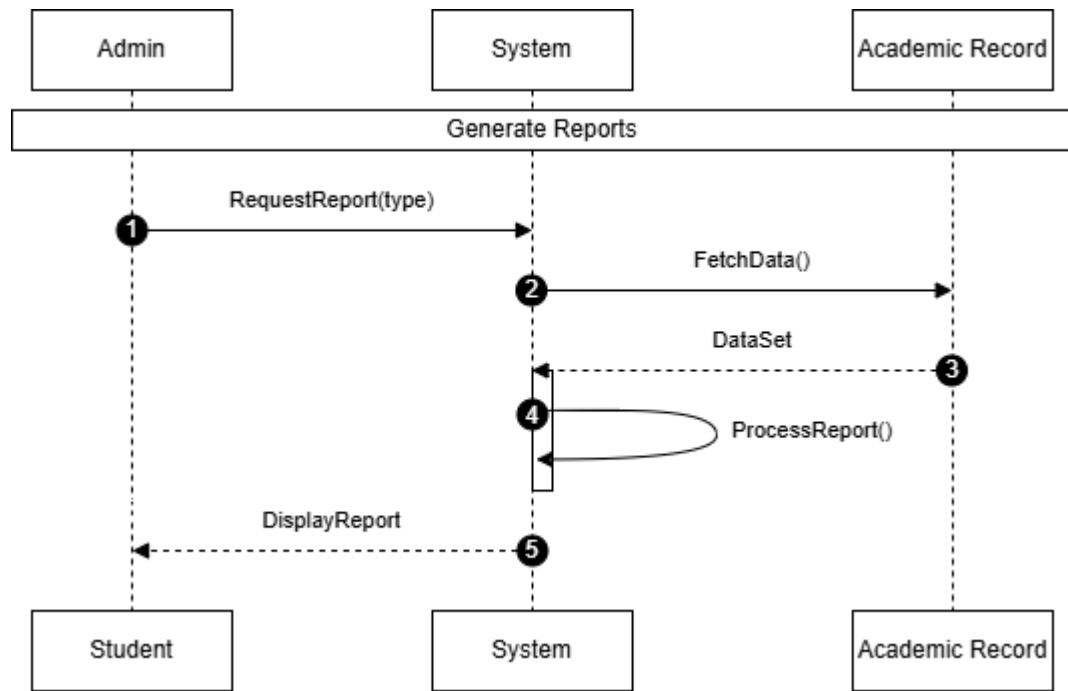


Figure 8.5 Sequence Diagram for Admin (Generating Reports)

This sequence diagram illustrates the process of generating reports in the Automated Student Records Management System (ASRMS). The interaction starts when the administrator requests a specific type of report, such as student summaries or verification status reports.

The system retrieves the necessary data from the database and processes it according to the requested format. Once the data has been compiled and organized, the system generates the report and presents it to the administrator. This process enables efficient data analysis and supports decision-making by providing summarized and structured information derived from the system's stored records.

3.8.4. Activity Diagram

Activity diagrams are used to illustrate the flow of processes and interactions within the proposed Automated Student Records Management System (ASRMS). These diagrams provide a visual

representation of how tasks, decisions, and actions are carried out by different users of the system, specifically the student, registrar, admin, and the system itself.

The diagrams highlight the step-by-step workflows involved in key system functionalities, including student record viewing, digital submission of personal information updates and registrar-managed record operations such as creating, updating, and deleting records. Each activity diagram is organized using swimlanes to clearly define the roles and responsibilities of each participant, ensuring a structured and role-based process flow.

Through these visual representations, the system's operational logic becomes easier to understand, particularly in identifying decision points, parallel processes, and data flow. The activity diagrams also emphasize the integration of validation, approval mechanisms, and activity logging, which are essential for maintaining data accuracy, security, and accountability within the system.

Overall, these diagrams serve as a guide in analyzing, designing, and communicating the functional behavior of the ASRMS, ensuring that all processes are efficient, consistent, and aligned with the system's objectives.

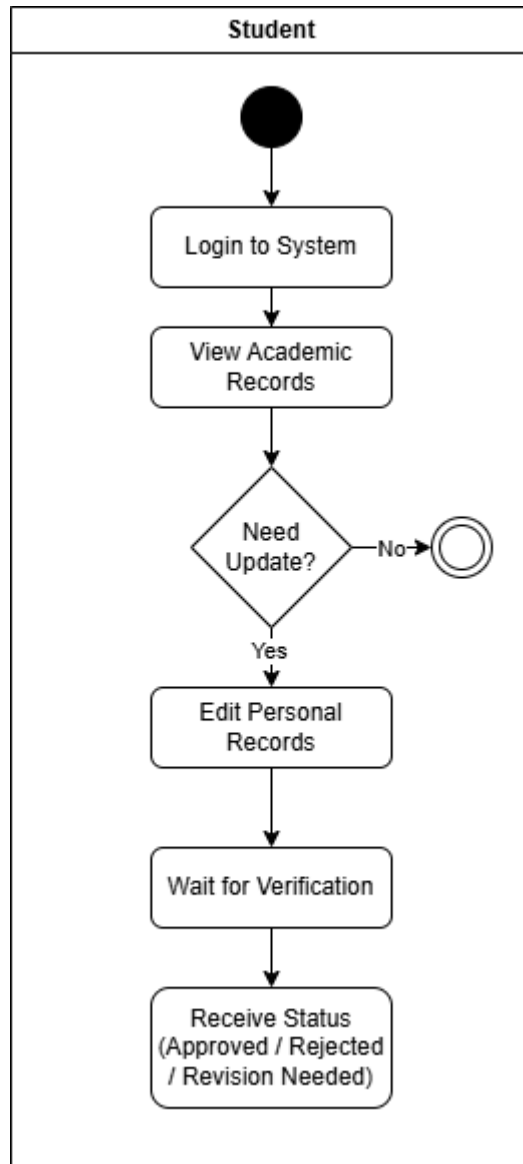


Figure 9.1 Activity Diagram for Student (Update Request)

The activity diagram illustrates the process followed by a student when requesting an update to their academic or personal records within the Automated Student Records Management System (ASRMS). The process begins when the student logs into the system and accesses their academic records for review.

After viewing the records, the student evaluates whether an update is necessary. If no changes are needed, the process terminates. However, if the student identifies incorrect or outdated information, they proceed to edit their personal records by entering the necessary modifications.

Once the changes are submitted, the student enters a waiting phase while the system forwards the request for verification. During this stage, the submitted updates are not immediately applied to the official record but are instead subject to validation and approval by the registrar.

Finally, the student receives a status notification indicating the result of the request, which may be approved, rejected, or returned for revision. The process concludes after the student is informed of the outcome. This workflow ensures that all updates undergo proper verification, maintaining the accuracy and integrity of student records.

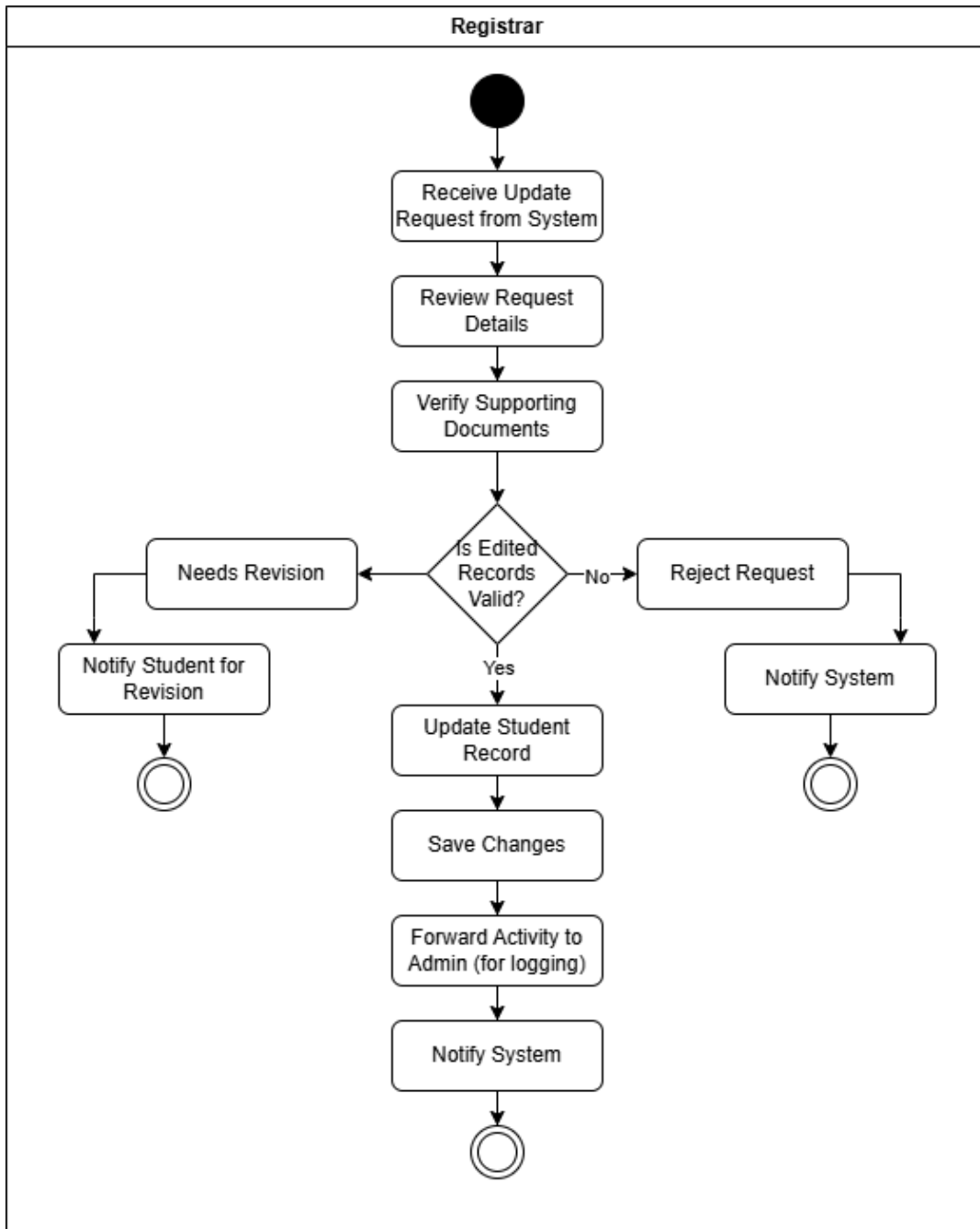


Figure 9.2 Activity Diagram for Registrar (Update Processing)

This activity diagram represents the workflow performed by the registrar in handling student update requests within the ASRMS. The process begins when the registrar receives a submitted request from the system. The registrar reviews the request details and verifies the supporting documents to ensure the accuracy and validity of the information provided.

A decision point follows, where the registrar determines whether the request is valid, requires

revision, or should be rejected. If the request is invalid, it is rejected and sent back through the system with appropriate feedback. If revisions are needed, the request is returned to the student for correction. If the request is approved, the registrar proceeds to update the student's academic record and saves the changes in the system.

After updating the record, the registrar forwards the activity to the admin module for logging and monitoring purposes. The system then notifies the student of the outcome. This process ensures that all record modifications are verified, controlled, and properly documented, maintaining the integrity of the database.

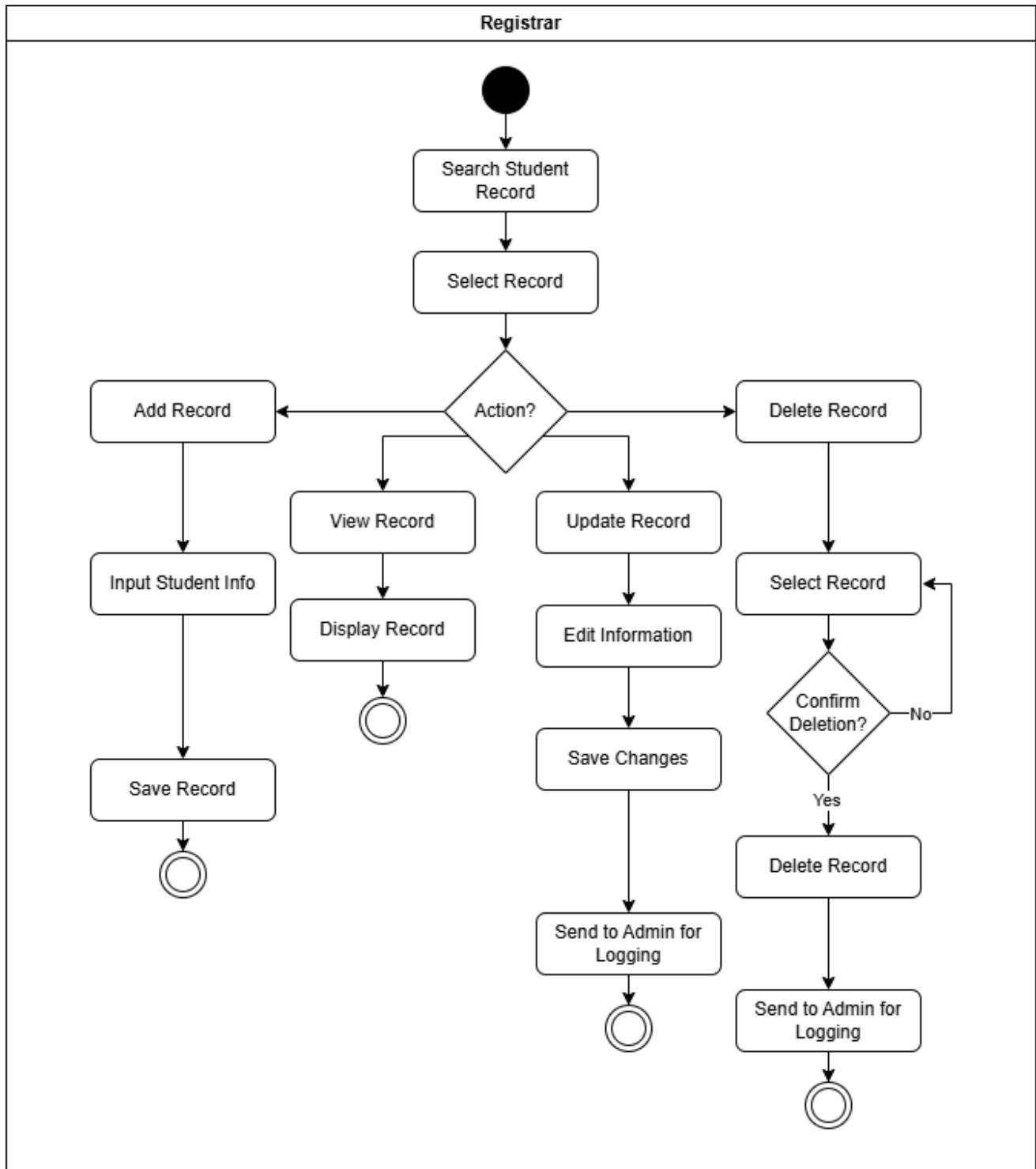


Figure 9.3 Activity Diagram for Registrar (Student Records Management – CRUD Operations)

This activity diagram illustrates the registrar’s role in managing student records through Create, Read, Update, and Delete (CRUD) operations within the ASRMS. The process starts when the registrar searches for a specific student record in the system and selects the desired record for further

action.

A decision point determines the type of operation to be performed. If the registrar chooses to add a new record, they input the necessary student information and save it in the system. If the action is to view a record, the system simply displays the relevant information without making any modifications.

For updating records, the registrar edits the necessary details and saves the changes, after which the system records the activity for administrative monitoring. In the case of deletion, the registrar must confirm the action before the system permanently removes the record. If the deletion is not confirmed, the process returns to record selection. Once confirmed, the record is deleted and the activity is logged by the admin. This workflow highlights controlled access to student data and ensures that all significant operations are tracked, promoting accountability, data accuracy, and system security.

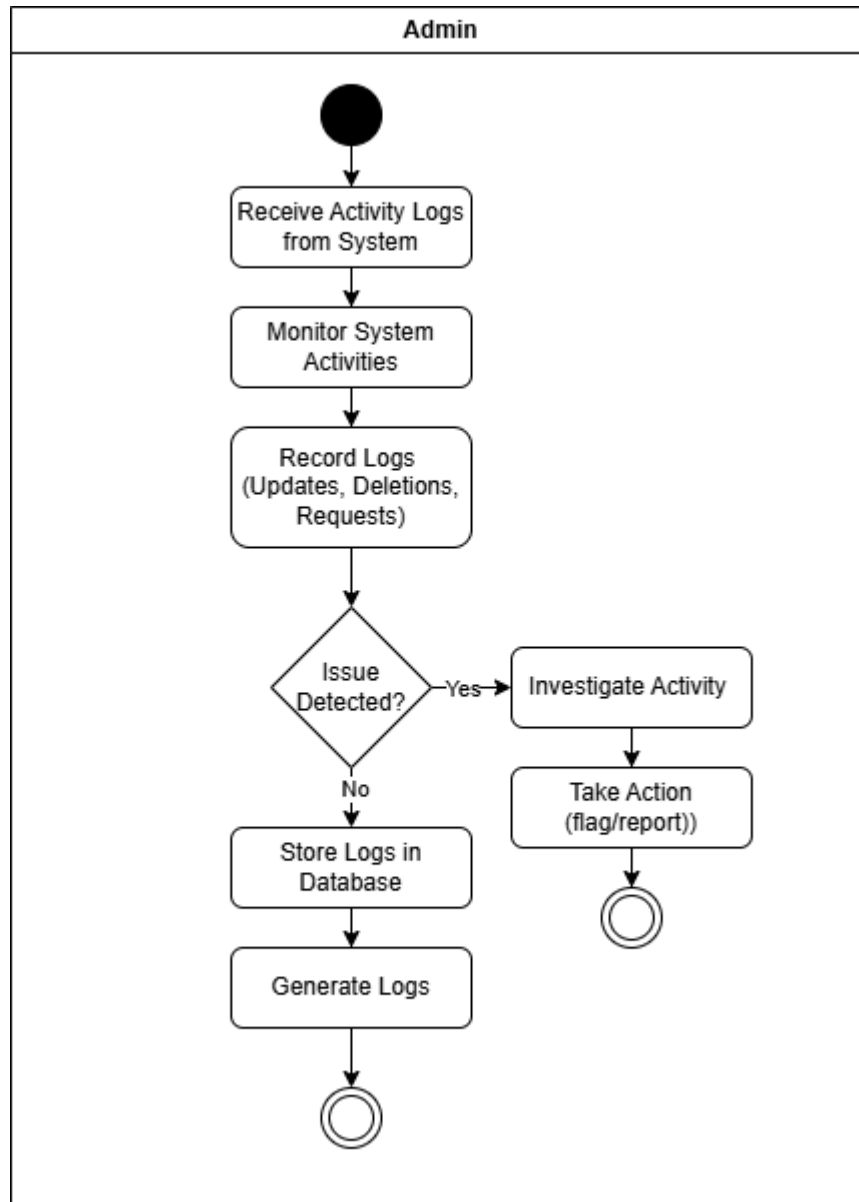


Figure 9.4 Activity Diagram for Admin (System Monitoring and Logging)

This activity diagram presents the responsibilities of the admin in monitoring and maintaining system activities within the ASRMS. The process begins when the admin receives activity logs generated by the system, particularly from registrar actions such as record updates, deletions, and approved requests.

The admin continuously monitors these activities to ensure that all operations are properly recorded and comply with system policies. The logs are then stored in the database for future reference,

auditing, and reporting purposes.

An optional decision point is included to address potential issues or anomalies detected within the system. If any irregularities are identified, the admin investigates the activity and takes appropriate action, such as flagging or reporting the issue. If no issues are found, the process proceeds with normal log storage. This workflow emphasizes the importance of transparency, accountability, and security by ensuring that all system activities are tracked and properly managed.

3.8.5.Statechart Diagram

The statechart diagram provides a visual representation of the lifecycle of a student record within the Automated Student Records Management System (ASRMS). It illustrates how a record moves through a series of defined states based on actions performed by the registrar, from its initial creation up to its final deletion. Each state reflects the current condition of the record, while the transitions indicate the operations or events that trigger movement from one state to another.

This diagram serves as a guide to understanding how student records are managed within the system, emphasizing the flow of activities such as validation, updating, archiving, and removal. By clearly mapping out these states and transitions, the statechart helps ensure that record handling follows a consistent and controlled process. It also highlights important decision points and possible loops, particularly during record updates and restoration, which contribute to maintaining data accuracy and integrity throughout the record's lifecycle.

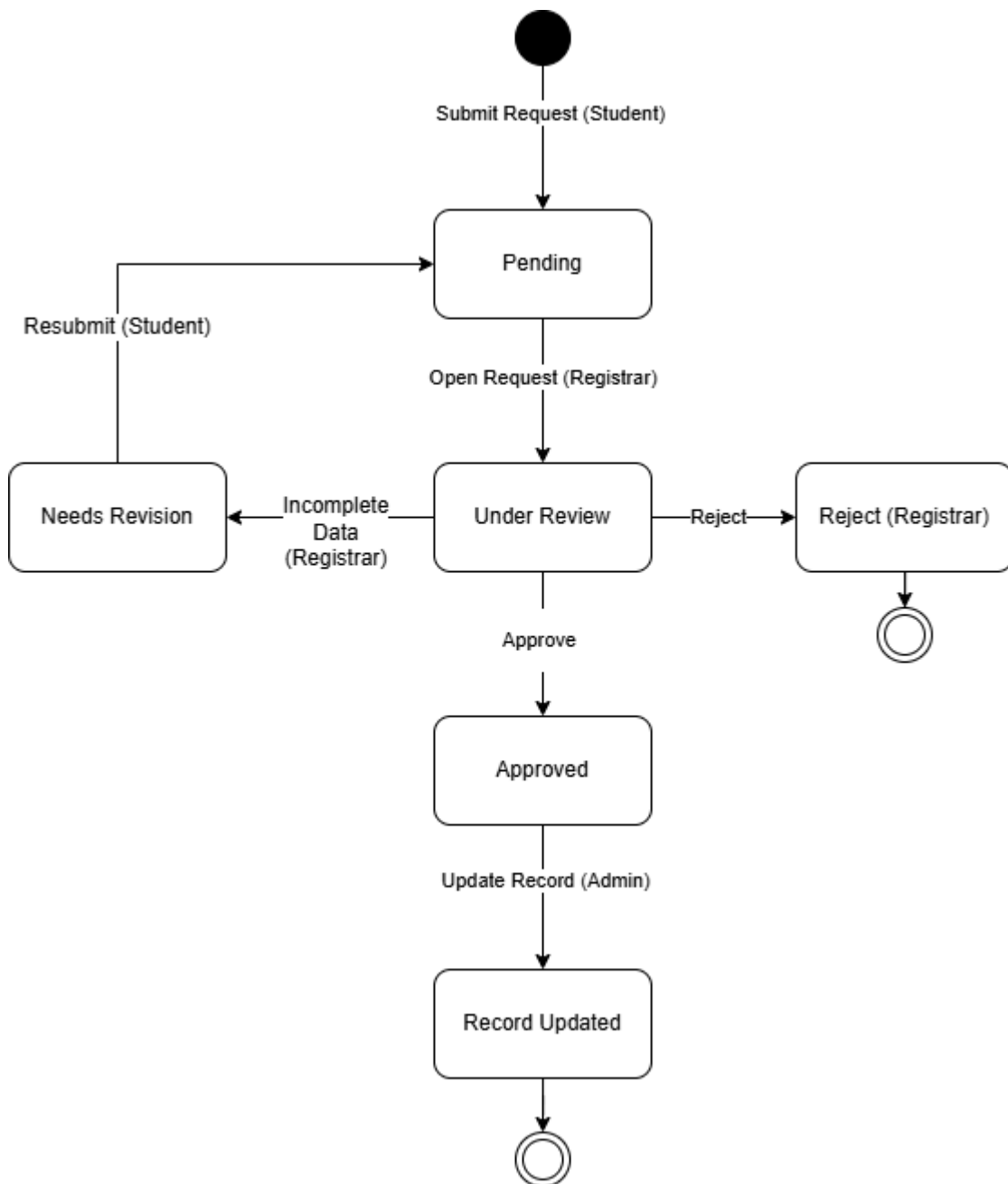


Figure 10.1 Statechart Diagram of Update Request Life Cycle in ASRMS

The statechart diagram illustrates the different states involved in processing a student record update request within the Automated Student Records Management System (ASRMS). It shows how a request transitions from one state to another based on actions performed by the student, registrar, and admin.

The process begins when the student submits a request, placing it in the **Pending** state. In this

state, the request awaits action from the registrar. Once the registrar opens the request, it transitions to the **Under Review** state, where the submitted information is evaluated.

During the review process, if the registrar identifies incomplete or incorrect data, the request moves to the **Needs Revision** state. The student is then required to correct and resubmit the request, which returns it to the **Pending** state for another review cycle.

If the registrar determines that the request is not valid or cannot be processed, it transitions to the **Rejected** state, ending the workflow. On the other hand, if the request meets all requirements, it is moved to the **Approved** state. Once approved, the admin proceeds to update the student's record in the system, transitioning the request to the **Record Updated** state. This marks the successful completion of the process.

This statechart emphasizes the lifecycle of a request, highlighting validation loops, decision points, and role-based actions that ensure accuracy, accountability, and proper handling of student record updates within the system.

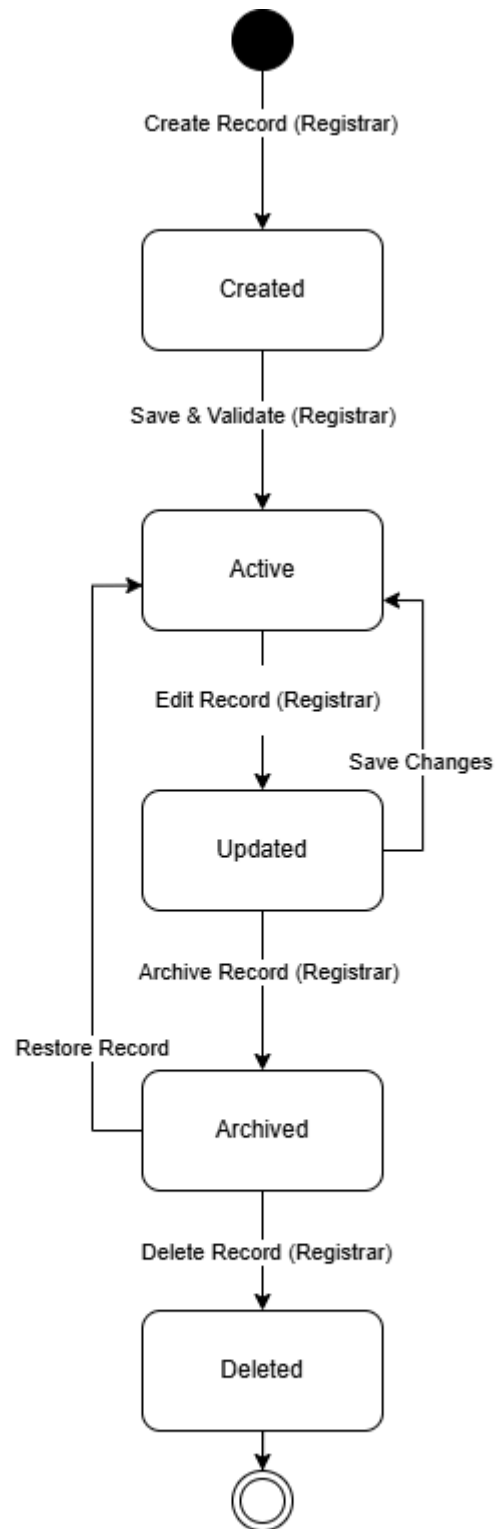


Figure 10.2 Statechart Diagram of Student Record Life Cycle in ASRMS

The statechart diagram illustrates the lifecycle of a student record within the Automated Student

Records Management System (ASRMS), detailing how a record transitions through various states based on actions performed by the registrar. It highlights the control flow, update cycles, and final disposition of records, ensuring proper data management and traceability throughout the system.

The process begins when the registrar creates a new student record, which is then placed in the Created state. At this stage, the record has been initialized but is not yet fully usable within the system. The registrar must perform a save and validation process to ensure that all required fields are complete and accurate. Once this validation is successful, the record transitions into the Active state.

In the Active state, the student record is considered valid and operational. It can be accessed, viewed, and modified as needed. If updates are required, such as correcting personal information or modifying academic details, the registrar initiates an edit action, moving the record into the Updated state. This state represents that changes have been made but are not yet finalized. After saving the changes, the record transitions back to the Active state, forming a continuous loop between Active and Updated. This loop ensures that records can be repeatedly maintained and improved while remaining usable.

When a record is no longer actively needed, such as when a student graduates, transfers, or becomes inactive, the registrar may archive it. This action transitions the record into the Archived state. In this state, the record is preserved for historical and reference purposes but is no longer part of the active dataset. However, the system allows flexibility; if needed, an archived record can be restored, returning it to the Active state and making it accessible again.

If the record is deemed unnecessary or must be permanently removed, due to policy, duplication, or data cleanup, it proceeds from the Archived state to the Deleted state through a delete action

performed by the registrar. The Deleted state represents the final stage in the lifecycle, where the record is no longer accessible within the system, marking the end of its existence.

Overall, this statechart emphasizes a structured lifecycle with clear transitions, including validation, update loops, archival processes, and final deletion. It ensures data integrity, controlled access, and proper record management by defining how records evolve over time within ASRMS.

3.8.6. Entity Relationship Diagram

The Entity-Relationship Diagram (ERD) shows the data structure of the Automated Student Records Management System and how key entities are connected. Main entities include Student, RecordTransaction, Staff, and Report. The Student entity contains personal and academic details, while RecordTransaction tracks requests, updates, and verifications handled by Staff. The Report entity compiles data from Student and RecordTransaction for monitoring and analysis. These entities are linked through primary and foreign keys to maintain data accuracy and consistency.

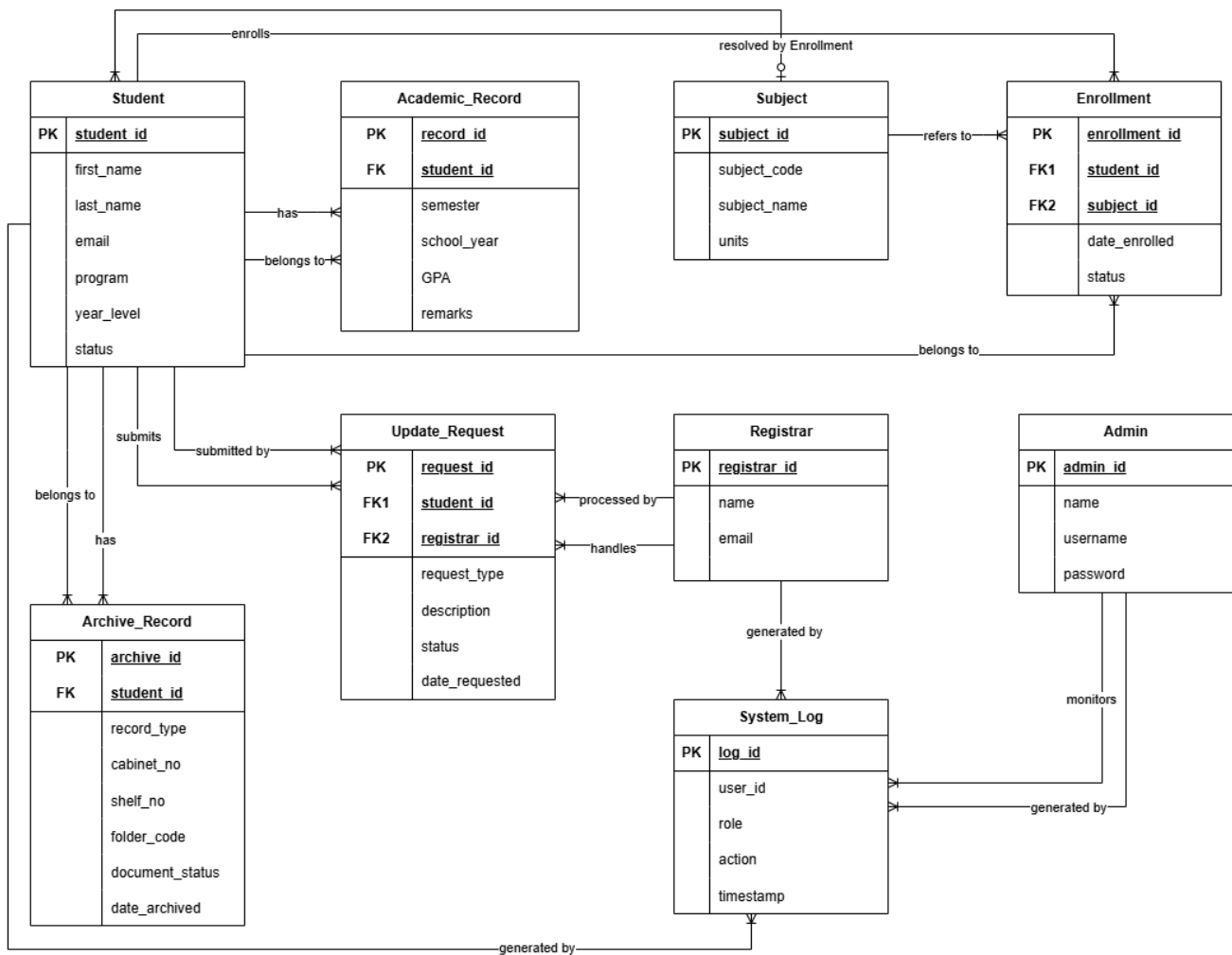


Figure 11 Entity Relationship Diagram for Student Management System

The diagram presents the Entity-Relationship structure of the proposed Automated Student Records Management System (ASRMS), illustrating how core data entities interact to support student record processing, enrollment, and administrative workflows. At the center of the system is the Student entity, which stores essential personal and academic information such as name, email, program, year level, and status. Each student is associated with an Academic_Record, which captures semester-based performance data including school year, GPA, and remarks, establishing a one-to-many relationship as a student may have multiple academic records over time.

Enrollment processes are represented through the Enrollment entity, which serves as a junction between Student and Subject, allowing students to enroll in multiple subjects while each subject can also be taken by multiple students. The Subject entity defines course-related details such as subject code, name, and units. This relationship ensures proper tracking of academic load and enrollment history, including enrollment date and status.

The system also incorporates a request management component through the Update_Request entity, where students can submit requests for record updates or corrections. These requests are processed and handled by the Registrar, establishing accountability and workflow traceability. Each request includes details such as request type, description, status, and date requested, enabling systematic monitoring and resolution.

For record retention and document management, the Archive_Record entity is linked to the student, storing information about archived files including storage location identifiers like cabinet number, shelf number, and folder code, along with document status and archival date. This ensures organized physical or digital record keeping.

Administrative oversight is represented by the Admin entity, which monitors system activities. Both registrar and admin actions are logged in the System_Log entity, capturing user roles, actions performed, and timestamps. This logging mechanism enhances transparency, security, and auditability within the system. Overall, the ER diagram demonstrates a well-structured relational database design that integrates student information management, enrollment tracking, request processing, archival storage, and system monitoring, ensuring efficient and reliable handling of academic records within the ASRMS.

3.8.7. Data Dictionary

The Data Dictionary provides a reference for developers, administrators, and stakeholders, defining entities, attributes, data types, and constraints in the Automated Student Records Management System. It ensures consistent data entry, validation, retrieval, and integrity, helping users and developers understand how information is stored and managed

Table 4. Data Dictionary

Module	Field Name	Data Type	Description	Constraints / Keys	Relationships
Users	id	BIGINT UNSIGNED	Unique user identifier	PK, NOT NULL, AUTO_INCREMENT	Referenced by Students, Staff, System_Logs
	name	VARCHAR (255)	Full name of user	NOT NULL	—
	email	VARCHAR (255)	User email address	UNIQUE, NOT NULL	—
	username	VARCHAR (255)	Username for login	UNIQUE, NOT NULL	—
	role	VARCHAR	User role	NOT NULL	—

		(255)	(Admin, Staff, Student)		
	department	VARCHAR (100)	Assigned department	NULL	—
	status	VARCHAR (20)	Account status	DEFAULT 'active'	—
	password	VARCHAR (255)	Encrypted password	NOT NULL	—
	remember_tok en	VARCHAR (100)	Remember login token	NULL	—
	created_at	TIMESTA MP	Record creation timestamp	NULL	—
	updated_at	TIMESTA MP	Record update timestamp	NULL	—
Students	student_id	BIGINT UNSIGNED	Unique student identifier	PK, NOT NULL, AUTO_INCR	Referenced by Enrollments, Grades,

				EMENT	Archive_Records, Program_Mappi ngs
	user_id	BIGINT UNSIGNED	Linked user account	FK, NULL	References Users(id)
	student_number	VARCHAR (20)	Official student number	UNIQUE, NOT NULL	—
	first_name	VARCHAR (50)	Student first name	NOT NULL	—
	middle_name	VARCHAR (50)	Student middle name	NULL	—
	last_name	VARCHAR (50)	Student last name	NOT NULL	—
	date_of_birth	DATE	Student birth date	NOT NULL	—
	email	VARCHAR (100)	Student email address	UNIQUE, NOT NULL	—

	contact_number	VARCHAR (15)	Contact number	NULL	—
	address	VARCHAR (150)	Home address	NULL	—
	place_of_birth	VARCHAR (120)	Birthplace	NULL	—
	sex	VARCHAR (10)	Gender/Sex	NULL	—
	guardian_name	VARCHAR (120)	Guardian full name	NULL	—
	citizenship	VARCHAR (60)	Citizenship	NULL	—
	elementary_school	VARCHAR (150)	Elementary school attended	NULL	—
	elementary_year	SMALLINT UNSIGNED	Elementary graduation year	NULL	—

	high_school	VARCHAR (150)	High school attended	NULL	—
	high_school_year	SMALLINT UNSIGNED	High school graduation year	NULL	—
	previous_school	VARCHAR (150)	Previous school attended	NULL	—
	previous_course	VARCHAR (150)	Previous course/program	NULL	—
	enrollment_date	DATE	Student enrollment date	NOT NULL	—
	graduation_date	DATE	Student graduation date	NULL	—
	GPA	DECIMAL(3,2)	Grade Point Average	NULL	—

	program_id	BIGINT UNSIGNED	Assigned academic program	FK, NULL	References Programs(id)
	created_at	TIMESTAMP	Record creation timestamp	NULL	—
	updated_at	TIMESTAMP	Record update timestamp	NULL	—
Programs	id	BIGINT UNSIGNED	Unique program identifier	PK, NOT NULL, AUTO_INCREMENT	Referenced by Students, Curriculum, Program_Mappings
	code	VARCHAR (20)	Program code	UNIQUE, NOT NULL	—
	name	VARCHAR (150)	Program name	NOT NULL	—
	description	VARCHAR (255)	Program description	NULL	—

	created_at	TIMESTAMP	Record creation timestamp	NULL	—
	updated_at	TIMESTAMP	Record update timestamp	NULL	—
Subjects	id	BIGINT UNSIGNED	Unique subject identifier	PK, NOT NULL, AUTO_INCREMENT	Referenced by Enrollments, Grades, Curriculum
	code	VARCHAR(20)	Subject code	UNIQUE, NOT NULL	—
	title	VARCHAR(150)	Subject title	NOT NULL	—
	units	TINYINT UNSIGNED	Credit units	NOT NULL, DEFAULT 3	—
	description	VARCHAR(255)	Subject description	NULL	—
	created_at	TIMESTAMP	Record	NULL	—

		MP	creation timestamp		
	updated_at	TIMESTA MP	Record update timestamp	NULL	—
Enrollments	id	BIGINT UNSIGNED	Unique enrollment identifier	PK, NOT NULL, AUTO_INCREMENT	—
	student_id	BIGINT UNSIGNED	Linked student	FK, NOT NULL	References Students(student _id)
	subject_id	BIGINT UNSIGNED	Linked subject	FK, NOT NULL	References Subjects(id)
	academic_year	VARCHAR (20)	Academic year	NOT NULL	—
	semester	VARCHAR (20)	Semester enrolled	NOT NULL	—
	status	VARCHAR	Enrollment	NOT NULL,	—

		(20)	status	DEFAULT 'enrolled'	
	created_at	TIMESTAMP	Record creation timestamp	NULL	—
	updated_at	TIMESTAMP	Record update timestamp	NULL	—
Grades	id	BIGINT UNSIGNED	Unique grade identifier	PK, NOT NULL, AUTO_INCREMENT	—
	student_id	BIGINT UNSIGNED	Linked student	FK, NOT NULL	References Students(student_id)
	subject_id	BIGINT UNSIGNED	Linked subject	FK, NOT NULL	References Subjects(id)
	academic_year	VARCHAR(20)	Academic year	NOT NULL	—

	semester	VARCHAR (20)	Semester	NOT NULL	—
	grade_value	DECIMAL(4,2)	Student grade	NULL	—
	remarks	VARCHAR (50)	Grade remarks	NULL	—
	created_at	TIMESTA MP	Record creation timestamp	NULL	—
	updated_at	TIMESTA MP	Record update timestamp	NULL	—
Curriculum	id	BIGINT UNSIGNED	Unique curriculum identifier	PK, NOT NULL, AUTO_INCREMENT	—
	program_id	BIGINT UNSIGNED	Linked academic program	FK, NOT NULL	References Programs(id)
	subject_id	BIGINT	Linked	FK, NOT	References

		UNSIGNED	subject	NULL	Subjects(id)
	year_level	TINYINT UNSIGNED	Assigned year level	NOT NULL	—
	semester	VARCHAR (20)	Semester offering	NOT NULL	—
	prerequisite	BIGINT UNSIGNED	Required prerequisite subject	FK, NULL	References Subjects(id)
	created_at	TIMESTAMP	Record creation timestamp	NULL	—
	updated_at	TIMESTAMP	Record update timestamp	NULL	—
Program_Mappings	id	BIGINT UNSIGNED	Unique mapping identifier	PK, NOT NULL, AUTO_INCREMENT	—

	program_id	BIGINT UNSIGNED	Linked program	FK, NOT NULL	References Programs(id)
	student_id	BIGINT UNSIGNED	Linked student	FK, NOT NULL	References Students(student_id)
	academic_year	VARCHAR (255)	Academic year	NOT NULL	—
	semester	VARCHAR (255)	Semester	NOT NULL	—
	status	VARCHAR (255)	Enrollment status	DEFAULT 'enrolled'	—
	year_level	INT	Student year level	NULL	—
	created_at	TIMESTAMP	Record creation timestamp	NULL	—
	updated_at	TIMESTAMP	Record update timestamp	NULL	—

Archive_Records	archive_id	BIGINT UNSIGNED	Unique archive identifier	PK, NOT NULL, AUTO_INCREMENT	—
	student_id	BIGINT UNSIGNED	Linked student	FK, NOT NULL	References Students(student _id)
	record_type	VARCHAR (255)	Type of archived record	NOT NULL	—
	cabinet_no	VARCHAR (255)	Cabinet number	NOT NULL	—
	shelf_no	VARCHAR (255)	Shelf number	NOT NULL	—
	folder_code	VARCHAR (255)	Folder identifier	NOT NULL	—
	document_status	VARCHAR (255)	Document status	NOT NULL	—
	created_at	TIMESTAMP	Record creation	NULL	—

			timestamp		
	updated_at	TIMESTAMP	Record update timestamp	NULL	—
Staff	staff_id	BIGINT UNSIGNED	Unique staff identifier	PK, NOT NULL, AUTO_INCREMENT	—
	user_id	BIGINT UNSIGNED	Linked user account	FK, NULL	References Users(id)
	first_name	VARCHAR(50)	Staff first name	NOT NULL	—
	last_name	VARCHAR(50)	Staff last name	NOT NULL	—
	role	VARCHAR(50)	Staff role	NOT NULL	—
	email	VARCHAR(100)	Staff email address	UNIQUE, NOT NULL	—

	created_at	TIMESTAMP	Record creation timestamp	NULL	—
	updated_at	TIMESTAMP	Record update timestamp	NULL	—
Reports	report_id	BIGINT UNSIGNED	Unique report identifier	PK, NOT NULL, AUTO_INCREMENT	—
	report_type	VARCHAR (50)	Type of generated report	NOT NULL	—
	generated_date	DATETIME	Date generated	NOT NULL	—
	content	TEXT	Report content/details	NOT NULL	—
	created_at	TIMESTAMP	Record creation	NULL	—

			timestamp		
	updated_at	TIMESTAMP	Record update timestamp	NULL	—
System_Logs	log_id	BIGINT UNSIGNED	Unique log identifier	PK, NOT NULL, AUTO_INCREMENT	—
	action	VARCHAR(255)	Action performed	NOT NULL	—
	user_id	BIGINT UNSIGNED	User who performed action	FK, NOT NULL	References Users(id)
	role	VARCHAR(255)	User role	NOT NULL	—
	created_at	TIMESTAMP	Record creation timestamp	NULL	—
	updated_at	TIMESTAMP	Record update	NULL	—

			timestamp		
System_Settings	id	BIGINT UNSIGNED	Unique settings identifier	PK, NOT NULL, AUTO_INCREMENT	—
	key	VARCHAR (100)	System setting key	UNIQUE, NOT NULL	—
	value	TEXT	Setting value	NULL	—
	created_at	TIMESTAMP	Record creation timestamp	NULL	—
	updated_at	TIMESTAMP	Record update timestamp	NULL	—

The data dictionary of the Automated Student Records Management System (ASRMS) provides a comprehensive definition of all data elements used within the system. It serves as a reference that describes each module, including its corresponding fields, data types, descriptions, constraints, and relationships. This ensures consistency and clarity in how data is structured, stored, and managed across the system.

Each module in the data dictionary represents a specific component of the system, such as users, Students, Programs, Subject, Enrollments, Grades, Curriculum , Program Mapping, , Archive Records, Staff, and System Settings. The fields within each module define the attributes of the data, while the data types specify the kind of values that can be stored. Constraints and keys, such as primary keys and foreign keys, are included to enforce data integrity and establish relationships between different modules.

Additionally, the data dictionary highlights how different modules are interconnected. For example, the Student module is linked to multiple components such as Academic Records, Enrollment, Program Mapping and Archive Records, ensuring a structured and relational database design. The inclusion of archive-related metadata, such as cabinet number, shelf number, and folder code, further supports the system's ability to manage both digital and physical records efficiently. Overall, the data dictionary plays a critical role in guiding the development and implementation of the system by providing a clear and organized representation of the database structure, ensuring accuracy, consistency, and maintainability of student data.

3.9. Functional Requirements

The Automated Student Records Management System must automate core student records functions:

1. Student Functional Requirements

Students interact with the ASRMS primarily to access and manage their personal academic information in a secure manner. The functional requirements for the Student role include:

Login and Authentication

Secure login and logout using unique credentials.

Password recovery and profile management.

View Personal Records

-Access and view personal information, academic history, course registrations, grades, and academic status.

-Ensure read-only access to academic records until processed and approved.

-Download available documents uploaded by authorized personnel.

Request Updates to personal information

-Submit requests to update personal information (e.g., address, contact number)..

2. Registrar Staff Functional Requirements

Registrar Staff manage, verify, and approve student records, ensuring accuracy and compliance with institutional policies. Functional requirements include:

Login and Authentication

-Secure login/logout with staff credentials.

-Role-based access control to restrict unauthorized functions.

Manage Student Records (CRUD)

-Create, read, update, and delete student personal and academic records.

-Perform manual data entry and encoding of student records through authorized registrar staff account interfaces.

-Validate, approve, and update academic information before records become

accessible to students.

- Upload digital forms or files associated with student records.

- Organize records with metadata and specify storage locations for legal documents.

Approve or Reject Requests

- Review and approve/reject student update requests.

- Verify academic information

- Generate verification status reports.

Document Management

- Provide students with access to approved academic and personal records through the institutional LAN-based system.

- Maintain audit logs for all record modifications and actions.

- Upload and distribute academic documents to students within the institutional network.

Reporting and Analytics

- Generate reports on student academic performance, record updates, and system activity.

- Support administrative decision-making through basic reporting and monitoring.

3. Admin Functional Requirements

Administrators oversee system governance, security, and overall operations, ensuring all activities comply with institutional policies. Functional requirements include:

System Management

- Manage user accounts and roles (create, modify, delete).

- Configure system settings, permissions, and approval workflows.

System Oversight and Governance

- Oversee approval workflows and monitor audit logs for compliance.
- Manage metadata and storage locations for document compliance.

Security and Audit

- Monitor system security, enforce encryption, and password protection.
- Access and review audit trails for transparency and accountability.

Reporting and Communication

- Generate comprehensive system and student data summary reports.
- Oversee overall system activity and maintain operational integrity.

3.10. Non - Functional Requirements

The Automated Student Records Management System must meet the following quality standards:

1. Student Non-Functional Requirements

Students primarily interact with the system to view records and request updates or documents. Their non-functional requirements include:

Usability

- The system interface must be user-friendly and intuitive for students of varying technical skills.
- Students should be able to submit requests and track statuses with minimal training.

Performance

- The system should load academic and personal records within 3 seconds.

Security

- Student login credentials must be encrypted and securely stored.
- Students must access only their own records; unauthorized access to other students' data is prohibited.

Availability

- The system must be accessible during institutional network availability, with minimal downtime for maintenance.

2. Registrar Staff Non-Functional Requirements

Registrar Staff manage records and approvals. Their non-functional requirements focus on efficiency, security, and reliability:

Performance

- CRUD operations on student records should complete within 2 seconds.
- Batch processing of requests must not degrade system performance.

Reliability

- All record updates must be reliably stored with proper versioning.
- The system should provide automated backup and recovery to prevent data loss.

Security

- Staff credentials must support multi-factor authentication.
- Access must be restricted to authorized staff only, following role-based access

control.

Auditability

- All staff actions (record edits, approvals, deletions) must be logged for monitoring and accountability.
- Reports on verification status and audit trails must be readily available.

3. Admin Non-Functional Requirements

Administrators ensure overall system governance, security, and monitoring. Their non-functional requirements include:

Performance and Scalability

- The system must support simultaneous access by multiple staff and administrators without performance degradation.
- Admin tasks, such as role management or full system reports, must execute efficiently (<5 seconds per operation).

Security and Privacy

- Admin access must be strictly controlled and protected by encryption and multi-factor authentication.
- All sensitive student data must comply with data privacy standards (e.g., secure storage, encrypted backups).

Reliability and Availability

- The system must maintain high availability ($\geq 99\%$) to ensure continuous administrative operations.
- Automated backups must occur daily, with disaster recovery procedures in place.

Maintainability

- Admins must be able to update system configurations and manage workflows without downtime.
- System updates should be logged and reversible in case of errors.

3.11. Project Details

3.11.1. Project Development

The project will utilize the Iterative and Incremental Software Development Model because it will provide an effective and structured approach in developing the TMCC Student Records Management System while allowing continuous feedback from system users. This development model will be appropriate for the registrar's office environment, where student record handling involves interconnected processes that require accuracy, security, and efficiency without interrupting daily operations.

The development process will be conducted through iterative and incremental cycles, beginning with the core module of student records management to address critical academic data storage, retrieval, and updating. Initial system versions will focus on automating the creation, maintenance, and organization of student academic records to reduce manual encoding errors and data redundancy (Pressman, 2020). Early testing phases are expected to reveal challenges such as inconsistent data entry, delayed document retrieval, and inefficient manual tracking of student records. Feedback from registrar personnel will guide improvements in system accuracy, accessibility, and usability.

Subsequent iterations will enhance record verification, approval workflows, and student records request processing. These improvements will ensure that submitted record updates and

document requests comply with institutional policies and maintain data integrity. Additional incremental features will include automated transcript generation, digital certification requests, transaction tracking, and notification systems, which will improve service efficiency and reduce processing time for student record requests. The phased release of system features will allow staff to gradually adapt to the new system while minimizing operational risks (Sommerville, 2021).

Overall, the iterative and incremental development approach will enable the continuous refinement of the Student Records Management System based on actual user requirements. TMCC will benefit from early access to functional record management tools, improved data accuracy, and a smoother transition from manual to automated record handling processes. The developed system will effectively support the registrar's operational needs and enhance the efficiency of student academic record management.

3.11.1.1. Planning

The planning phase will establish the project's foundation by examining TMCC's academic record-keeping and administrative workflows. The review will identify operational inefficiencies caused by manual data entry, fragmented student databases, and repetitive paperwork in transcript processing, grade encoding, and student file maintenance. Before development begins, TMCC will conduct its own feasibility study to evaluate technical readiness and resource capacity for a centralized automated system. The results are expected to confirm that the college has sufficient infrastructure, committed leadership, and qualified personnel to support the automated student records management system. The study will also identify priority record-handling processes for automation, which will shape the project's functional scope and its goal of improving institutional efficiency.

Guided by these findings, the development team will hold consultations with registrars, admissions officers, IT staff, and faculty to verify operational bottlenecks and refine user requirements for record management. From these discussions, the project objective will be defined: to build an integrated automated student records management system that strengthens data accuracy, speeds up record retrieval and document issuance, and improves coordination across offices. A work breakdown structure and timeline will then be prepared to organize tasks, clarify responsibilities, and align the system's implementation with TMCC's institutional targets for operational excellence.

3.11.1.2. Requirement Analysis

The requirements analysis phase will aim to identify the specific needs of TMCC's student record stakeholders and map institutional data flows for system design. Data will be collected through interviews, surveys, and direct observation within the Registrar's Office and Admissions Department. Functional requirements, such as viewing, downloading, and updating personal records, as well as CRUD operations for the registrar, will be documented alongside non-functional requirements, including AES-standard data security and efficient system response times. User stories and detailed data flowcharts will identify operational bottlenecks, such as the manual verification of student credentials, and define desired system capabilities, including secure access and efficient record management. Constraints such as legacy database compatibility and compliance with data privacy regulations will also be identified to guide feasible and effective design choices. The outcome of this phase will be a comprehensive requirements specification document that will serve as a technical blueprint for development, ensuring the automated student records management system effectively eliminates operational delays and meets TMCC's institutional needs.

3.11.1.3. System Design

The system design phase will translate the project requirements into a functional architecture that organizes the Automated Student Record Management System (ASRMS) into clear and modular components. The design groups the system into layers that separate data management, system processes, and user interaction to improve maintainability, organization, and operational stability. Each module is designed to support the core operational needs identified during the requirements analysis phase, particularly in improving the accuracy, accessibility, and management of student academic and personal records.

The Records Management Module will serve as the centralized repository of student academic and personal information, allowing authorized Registrar personnel to efficiently manage student profile information, grades, and related academic records within the institution. The Request Management Module will support the submission and monitoring of student information correction requests within the institution's LAN-based environment. Students may request updates or corrections to personal information, subject to verification and approval by authorized Registrar personnel.

The system will also include validation and approval mechanisms to ensure that only verified and authorized changes are applied to the database. Access control features, audit trails, and security configurations will be integrated into the system to help protect sensitive academic and personal information and monitor system activities performed by authorized users.

The design utilizes institutional data stores such as Student Records, Transaction Logs, and Course Information to support organized data handling and reduce duplication of records. By structuring the system in this manner, the ASRMS aims to improve record retrieval, monitoring, and

management processes within the Registrar's Office while maintaining data consistency and operational efficiency within the institution's internal network environment.

3.11.1.4. Coding

The coding phase will involve converting the system design into a fully functional Automated Student Record Management System (ASRMS). The development team will employ Laravel as the PHP framework for back-end processing, MySQL for secure database management of academic records, and React JS as a JavaScript library to enhance front-end interactivity for administrative staff and students. Strict coding standards will be observed to ensure data integrity, scalability, and long-term maintainability. Version control with Git will enable collaborative development, allowing the team to track changes in the record-processing logic and recover from potential errors during the build.

Each core component, including Records Management, Verification Logic, Request Processing, and Reporting, will be developed incrementally. This modular approach will allow continuous testing and debugging, specifically focusing on the accuracy of data migration from TMCC's legacy files. Regular coordination with TMCC's Registrar and IT staff will ensure that each module meets the specific operational requirements for record retrieval and document issuance. By aligning the code with actual institutional workflows, the development team will optimize system usability and ensure that the automated platform directly enhances the operational efficiency of student record handling.

3.11.1.5. Testing

The testing phase will verify that the Automated Student Records Management System (ASRMS) effectively meets the study's objectives, particularly in improving operational efficiency through process automation in student record management. The system will be evaluated across multiple testing levels to ensure that all core functionalities perform according to the specified requirements.

Unit testing will be conducted on each major module of the system, including Student Records Management, Update Request Processing, Archive Record Management, and Logs and Reporting. This phase will ensure that individual features such as student data entry, record updating, request submission, and retrieval of archived records—including cabinet location metadata—function accurately and reliably.

Following this, integration testing will be performed to validate the interaction between different system components. This will include verifying the seamless data exchange between the student records database, update request module, and archive records repository. The testing will ensure that information flows correctly across modules, particularly when retrieving both digital records and physical file locations, without errors or data inconsistencies.

User Acceptance Testing (UAT) will involve registrars and administrative staff executing real-world tasks within the system. These tasks will include processing update requests, retrieving student academic records, locating physical documents using cabinet metadata, and generating reports. This phase will focus on evaluating system usability, response time, and data accuracy. Feedback gathered from users will be used to refine the interface and improve alignment with the registrar's workflow.

Performance and reliability testing will assess the system's responsiveness, stability, and accuracy under varying workloads. The system will be tested to ensure it can handle multiple concurrent users and maintain performance during peak periods, such as enrollment and records processing. The results are expected to confirm that the system maintains consistent speed and reliability without performance degradation. Overall, the testing process will evaluate whether the ASRMS improves efficiency by reducing manual processes, minimizing data redundancy, and enabling faster access to both digital and physical student records. The system will provide a reliable and scalable solution that supports the institution's goal of modernizing and streamlining its record management operations.

3.11.1.6. Deployment

Deployment will commence with a pilot implementation in the Registrar's and Admissions offices to minimize risk and allow for workflow adjustments. The system will be installed on TMCC's secure servers with role-based access to protect sensitive student data. Training sessions and user manuals will facilitate a smooth transition from manual filing to the automated platform.

Following stable performance and positive feedback on retrieval speeds, full-scale deployment will be executed. This phase will include establishing a maintenance routine for backups and security patches to ensure long-term data integrity. The successful transition to the ASRMS will mark a key milestone in improving TMCC's operational efficiency and administrative accuracy.

3.11.1.7. Maintenance

Once deployed, continuous maintenance will become essential to ensuring the system's long-term

functionality and security. A structured plan will be developed to address both preventive actions, such as database optimization and server monitoring, and corrective actions, including bug fixes and troubleshooting reported by the Registrar's Office.

The development team will coordinate with TMCC's IT Unit to establish a maintenance plan for the Automated Student Records Management System (ASRMS). During deployment and turnover, monthly database backups will be recommended to ensure the continued protection of student records. A maintenance window of two to four hours during non-peak evening hours will also be suggested to minimize disruptions during system updates and routine checks.

A built-in user feedback module will be included to allow staff to report technical concerns directly to the system administrator, supporting continuous system monitoring and improvement after deployment. This feature will ensure that issues can be properly documented and addressed by TMCC's IT Unit as part of their system management responsibilities.

3.12. Evaluation Procedure

The evaluation phase assessed the system's effectiveness in achieving its core objectives, specifically determining how the Automated Student Record Management System (ASRMS) enhanced TMCC's operational efficiency. Quantitative metrics, such as the reduction in record processing time, improved data accuracy, and decreased error rates, were measured against legacy manual baselines. Qualitative feedback from surveys and focus group discussions with the Registrar and administrative staff provided insights into usability and reliability, guiding iterative refinements like interface improvements and workflow optimizations for document issuance.

System effectiveness was measured through key operational efficiency indicators, focusing

on the turnaround time for transcript requests and the precision of automated record retrieval. Data accuracy was tracked through the reduction of discrepancies in student archives, while stakeholder adoption was assessed via surveys to ensure the system met institutional goals. Continuous monitoring confirmed that the ASRMS provided a reliable, efficient, and user-centered solution that enhanced institutional service delivery and supported TMCC's digital transformation.

3.13. Software Quality Characteristics

The Automated Student Record Management System (ASRMS) for Trece Martires City College was developed following recognized software quality standards to ensure effectiveness, reliability, and long-term usability. The system's functionality was designed specifically to meet TMCC's academic record-keeping needs, automating the lifecycle of student data from enrollment verification to graduation. Each feature—including automated grade consolidation and digital transcript generation—was based on stakeholder requirements to ensure outputs remain accurate, consistent, and compliant with institutional policies. To guarantee reliability, the system underwent stress testing to ensure stability during high-traffic periods, such as grade encoding weeks, while built-in backup mechanisms and error-handling procedures were implemented to prevent data loss.

In terms of usability, the system features an intuitive and responsive interface that allows the Registrar's staff and students to navigate functions easily without extensive technical training. Operational efficiency was prioritized by optimizing database queries and processing speeds, which significantly reduced the turnaround time for record retrieval compared to previous manual methods. Maintainability was achieved through a modular structure and well-documented code,

allowing for future updates to the verification logic without disrupting the core database. The system's portability ensures compatibility across various devices and browsers, allowing for secure access whether on-premise or through cloud-based deployment.

Given the sensitivity of academic records, security protocols such as role-based access control (RBAC), data encryption, and secure authentication were implemented to ensure compliance with data privacy standards. The system also demonstrates strong compatibility, integrating smoothly with existing institutional tools and providing a scalable architecture to accommodate TMCC's growing student population without performance degradation. Collectively, these quality characteristics ensure that the ASRMS is a functional, secure, and sustainable solution that directly supports the institution's shift toward digital transformation and enhanced operational efficiency.

3.14. Projects Cost

The project cost will cover all expenses necessary for the successful development, testing, and deployment of the Automated Student Record Management System (ASRMS) for Trece Martires City College. Major cost components will be categorized into hardware, software, manpower, and operational expenses. Hardware costs will include secure servers and backup storage specifically sized to handle TMCC's academic archives, while software expenses will cover database licenses and secure cloud hosting for off-site data redundancy. Manpower costs will account for the development team—including system analysts and programmers—who will be dedicated to building the record-handling logic.

Operational expenses will include the infrastructure required to maintain a centralized records database. Additional funds will be allotted for specialized training sessions for the

Registrar’s staff and faculty to ensure a smooth transition to the automated platform, as well as recurring system maintenance to protect data integrity. A cost-benefit analysis will be conducted to justify the project’s financial feasibility, demonstrating how the long-term gains in operational efficiency, such as reduced man-hours and faster document issuance, will outweigh the initial investment.

The budget for system development and deployment will be managed through the institutional ICT allocation. This will ensure that every expenditure will contribute directly to the project’s objective of enhancing TMCC’s administrative efficiency. By monitoring budget allocation throughout the system development lifecycle, the college will maintain financial control while securing a sustainable automated solution for its growing student record management needs.

Table 5. Estimated Project Cost Breakdown

Category	Item Description	Estimated Cost (₱)
Hardware	A desktop computer will be used for development testing	20,000
Software	A database management system (MySQL/PostgreSQL setup) will be implemented.	3,000
Manpower	Two TMCC IT personnel will be assigned to the project.	N/A

Operational	Printing, documentation, and communication materials will be prepared and utilized.	500
Maintenance	Post-deployment debugging and security updates will be performed.	1,000
Total Estimated Project Cost		₱24, 500

The estimated project cost for the proposed system amounts to ₱24,500. The budget allocation covers the essential resources needed for the development, implementation, and maintenance of the system. A desktop computer worth ₱20,000 will be utilized for system development and testing purposes to ensure proper functionality and performance. An estimated ₱3,000 is allocated for software requirements, including the setup and configuration of the database management system such as MySQL or PostgreSQL.

In terms of manpower, two TMCC IT personnel will be assigned to participate in the development and implementation process without additional labor cost, as their services are institutionally provided. Operational expenses amounting to ₱500 will cover printing, documentation, and communication materials necessary throughout the project. Furthermore, ₱1,000 is allocated for maintenance activities, including post-deployment debugging, minor system fixes, and security updates to ensure the reliability and stability of the system after implementation.

3.15. Project Risk Management Plan

The Project Risk Management Plan will ensure that potential challenges during the development and implementation of the Automated Student Records Management System (ASRMS) for Trece Martires City College will be properly identified, monitored, and controlled. Considering the sensitivity and confidentiality of student information, the plan will focus on five major risk areas: technical failures, data security threats, schedule delays, budget constraints, and user resistance.

To address technical risks, regular code reviews, system backups, and version control using Git will be implemented to prevent data loss and maintain system reliability. Data security risks will be treated as the highest priority and will be mitigated through the implementation of encrypted data security measures, role-based access control (RBAC), and compliance with the Data Privacy Act of 2012 to ensure the protection of confidential student records.

Schedule-related risks will be minimized by following a structured project timeline supported by weekly progress monitoring and milestone evaluations, particularly focusing on the development of core student record modules. Budget risks will be controlled by adhering to the approved project allocation of ₱38,500, prioritizing essential database and system functionalities over secondary interface enhancements.

User resistance, which will be a common challenge in system implementation, will be addressed by involving the Registrar's Office and faculty members during early testing phases. Orientation and training sessions will be conducted to enhance user familiarity and confidence, ensuring a smoother transition from manual record management to an automated system.

Table 6 Project Risk Management Plan

Risk Category	Potential Issue	Probability	Impact	Planned Action
Technical	Server Downtime or Database Failure	Medium	High	Daily automated backups will be implemented to the external storage device, and cloud hosting with 99.9% uptime will be utilized.
Security	Unauthorized Access to Student Grades	Low	High	Role-Based Access Control (RBAC) and AES-256 data encryption will be implemented, and strong password policies will be enforced.
Data Integrity	Manual Entry Errors during Migration	High	Medium	Data validation scripts will be used to flag inconsistent formats, and a "dual-entry" verification process will be conducted during the pilot phase.
Operational	User Resistance from Registrar Staff	Medium	Medium	Hands-on training sessions will be conducted (as budgeted), and staff will be involved in the User Acceptance Testing (UAT).
Project	Schedule Delays (Missed	Medium	Medium	A strict Gantt chart will be followed, and core "Record Search"

	Milestones)			and "Encoding" modules will be prioritized over secondary features.
Financial	Budget Overrun for Hosting/Storage	Low	Low	The ₱48,500 budget will be adhered to, and open-source software will be utilized to eliminate licensing costs.

Table 6 will outline the Project Risk Management Plan for the Automated Student Records Management System (ASRMS) of Trece Martires City College. It will identify five major risks that may affect project delivery, each of which will be prioritized based on likelihood and impact, with corresponding mitigation and contingency actions.

Technical failure will be rated as a high-priority risk because the system's effectiveness will depend on stable software and hardware integration. To address this, the team will perform regular code reviews, implement backup protocols using the allocated external storage device, and enforce version control practices. These actions will aim to reduce system downtime and maintain the operational continuity required for seamless student record retrieval.

Data security threats will also be assigned high priority due to the sensitive nature of the academic records handled by the ASRMS. Encryption methods, secure login authentication through role-based access control (RBAC), and strict compliance with the Data Privacy Act of 2012 will be implemented to safeguard confidential information. Regular security audits within the cloud hosting environment will be conducted to help prevent unauthorized access and potential data breaches.

Time delays will be classified as a medium-priority risk. The team will implement milestone tracking

and weekly evaluations to ensure that the core system modules, including record search and grade encoding functionalities, will be completed according to schedule, thereby minimizing the risk of project overruns during peak academic periods.

Budget constraints will also be assessed as a medium-priority risk. Adherence to the approved project budget estimate of ₱48,500 through transparent cost monitoring and the use of open-source development tools will help ensure that the project will remain financially viable for Trece Martires City College.

User resistance will be considered a medium-priority risk but will remain a critical factor in ensuring successful system adoption. To mitigate this, the project team will involve the Registrar's Office and faculty members during the early stages of User Acceptance Testing (UAT). Training sessions and orientation activities, funded through the operational budget, will be conducted to build user familiarity and confidence, ensuring a smooth transition from manual record management to the automated system upon deployment.

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Survey Questionnaire

AUTOMATED STUDENT RECORD MANAGEMENT SYSTEM (ASRMS) SURVEY FOR TRECE MARTIRES CITY COLLEGE

This survey aims to assess the efficiency, accuracy, accessibility, communication, and satisfaction levels of users regarding the current record management processes and their readiness for the implementation of the ASRMS at Trece Martires City College.

Instructions: Please rate each statement based on your experience using the scale below.

Scale	Description
5	Always
4	Often
3	Sometimes
2	Rarely
1	Never

Part I. Respondents' Profile

Indicators	Options
1. Role Position	☐ Administrator ☐ Faculty ☐ Staff ☐ Student

2. Department Unit	☐ Registrar's Office ☐ Academic Department ☐ IT Unit ☐ Other: _____
3. Years of Service/Enrollment	☐ Less than 1 year ☐ 1–3 years ☐ 4–6 years ☐ 7 years and above
4. Frequency of System Use	☐ Daily ☐ Weekly ☐ Monthly ☐ Rarely
5. Computer Literacy Level	☐ Beginner ☐ Intermediate ☐ Advanced

Part II. Process Efficiency

Statements	5	4	3	2	1
1. Student record update handling is completed promptly.	☐	☐	☐	☐	☐
2. Enrollment-related record processing operates smoothly without delays.	☐	☐	☐	☐	☐
3. Manual record-keeping procedures often cause redundancy and errors. <i>(reverse item)</i>	☐	☐	☐	☐	☐
4. There is a clear, fast workflow for requesting and releasing academic documents.	☐	☐	☐	☐	☐
5. Automated coordination among departments minimizes repetitive data entry.	☐	☐	☐	☐	☐

Part III. Data Accuracy and Reliability

Statements	5	4	3	2	1
6. Student grades and personal records are consistently updated and accurate.	☐	☐	☐	☐	☐
7. Clerical errors in transcript or diploma preparation are minimal.	☐	☐	☐	☐	☐
8. Retrieval of archived student information is	☐	☐	☐	☐	☐

consistent and reliable.					
9. The current system ensures the secure handling of sensitive student data.	☐	☐	☐	☐	☐
10. Duplicate student IDs or missing records are common in the current system. (<i>reverse item</i>)	☐	☐	☐	☐	☐

Part IV. Accessibility and Usability

Statements	5	4	3	2	1
11. Access to academic records is easy and efficient for authorized users.	☐	☐	☐	☐	☐
12. The current interface for viewing or encoding student data is user-friendly.	☐	☐	☐	☐	☐
13. Students can easily access their own grades or status when needed.	☐	☐	☐	☐	☐
14. Sufficient technical support is available when using digital record tools.	☐	☐	☐	☐	☐
15. System access for record verification is often limited or inconsistent. <i>(reverse item)</i>	☐	☐	☐	☐	☐

Part V. Communication and Coordination

Statements	5	4	3	2	1
16. Record sharing between the Registrar and Academic departments is well-facilitated.	☐	☐	☐	☐	☐
17. Status updates on record update handling are effectively communicated to students.	☐	☐	☐	☐	☐
18. There is an efficient digital mechanism for addressing record discrepancies.	☐	☐	☐	☐	☐

19. Inter-departmental data synchronization supports fast decision-making.	☐	☐	☐	☐	☐
20. Miscommunication regarding student files causes significant process delays. <i>(reverse item)</i>	☐	☐	☐	☐	☐

Part VI. User Satisfaction and Improvement Readiness

Statements	5	4	3	2	1
21. Current record-keeping processes meet the needs of all stakeholders.	☐	☐	☐	☐	☐
22. Users are satisfied with the speed of obtaining academic certifications.	☐	☐	☐	☐	☐
23. The system supports effective monitoring of academic progress and history.	☐	☐	☐	☐	☐
24. I believe the ASRMS will significantly improve overall institutional operations.	☐	☐	☐	☐	☐
25. I am fully willing to adopt and use the new Automated Student Record Management System.	☐	☐	☐	☐	☐